2019-2025

AS BIOLOGY

asbiology

Notes from MS

Notes from MS

Notes from MS

MTK

MaiTK

IAS Biology U2 Notes From MS

R.D= Repeated Definition R.Q= Repeated Question

1. Cell Structure:

01. The organisation of cells "cells, tissues, organs and organ systems":

Term	Definition
(R.D): Tissue	*group of (similar) cells working together to perform a (specific) function*
Organ	*(group of) tissues which work together to perform {a function / functions}*
Organ system	*(a group of) organs that work together to perform {one function / more than one function / specific function / named function}*

- Give two differences between an organ and a tissue.
  - *organ has many functions and many / several / group of} {cell types / tissues}, whereas a tissue has {one / fewer} functions and {one cell type / similar cells}*

---

02. Ultrastructure of the Cell:

➔ Which organelles have or produce ribosomes? *(Nucleus and rER).*

- Name two structures that could be present inside the nucleus.
  - *nucleolus and DNA / chromatin / chromatid / chromosomes / nucleosome / mRNA.*

➔ Ribosomes are:
  - *located on the surface of rER (80s ribosomes on rER) and inside chloroplasts and mitochondria (70s Ribosomes)*

➔ A cell was supplied with radioactive amino acids. The cell took in these amino acids and used them in protein synthesis. Which structure in the cell would become radioactive first? *Ribosomes*.

- Name two structures present in animal cells that are **not** present in a plant cell.
  - *cilia, glycogen (granules), flagella, centrioles, lysosomes.*
- Suggest why mitochondria would be located within 10 nm of lipid droplets.
  - *it is a short distance between (mitochondria and lipid so quicker diffusion of {lipid / fatty acid / glycerol} (into mitochondria), and lipids are used in {respiration / ATP production} as the lipid droplets are converted to glucose.*

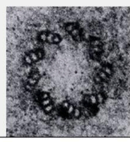
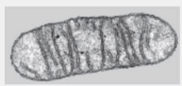
Structure	Function
Nucleus	• *DNA synthesis / DNA replication / semi-conservative replication* • *(site of) transcription / production of mRNA* Site of *post- transcriptional modification / (m)RNA splicing* • *site of epigenetic modification* • *(nucleolus) contains the genetic material.* • *controls the activities of the cell.* • *production of RNA / RNA nucleotides*
Nucleolus	• *{produce / assemble} ribosomes / produce rRNA / ribosomal subunits* • *synthesis of RNA*
Lysosome	• *contains {digestive enzymes / hydrolytic enzymes / lysozymes} that digest {pathogens / cells / components / organelles / cellular waste / glycogen}*
Centriole	• *form spindle {fibres / poles} in {prophase / mitosis / meiosis / cell division} to separate {chromosomes/ chromatids} (in cell division)*
Spindle Fibres	• *the {attachment/binding} of centromeres (to the spindle fibres) to allow the separation of the {(daughter) chromatids / chromosomes}, so that each daughter cell gets identical genetic material in mitosis*
Mitochondria	• *aerobic respiration to produce ATP needed for cell metabolism, mitosis, protein synthesis, muscle contraction, and active transport.* • *contains ribosomes for protein synthesis.*
Cell Membrane	• *controls what enters and leaves the cell.*
Golgi Apparatus	• *modifies and packages {proteins/enzymes}* • *forms extracellular enzymes* • *processes and packages lipids.*
Ribosomes	• *(site of) {translation / polypeptide synthesis}/ {produce / make} {polypeptides / 1° structure}/ form {chains / sequence} of amino acids/ translates mRNA base sequence into an amino acid sequence*
sER	• *synthesises of lipids / steroids.*
Membranes	• *are partially permeable and control what {enters/leaves} a certain structure e.g cell* • *{compartmentalise / encloses} the {DNA /ribosomes} from the cytoplasm* • *formation of vesicles and allows fusing/ exocytosis of vesicles to release proteins*

- Compare and contrast the structure and function of lysosomes and acrosomes.
  - *Similarities*
    - *both contain (digestive / hydrolytic) enzymes*
    - *both {have single membrane / membrane bound (organelles)}*
  - *Differences*
    - *a lysosome is spherical and involved with intracellular digestion, whereas an acrosome is {curved / not spherical} and involved with extracellular digestion.*
    - *a lysosome (enzyme) is involved in breaking down {cell components / microbe cells / virus} whereas an acrosome (enzyme) is involved in*

digesting the {outer layers of an egg cell / zona pellucida}

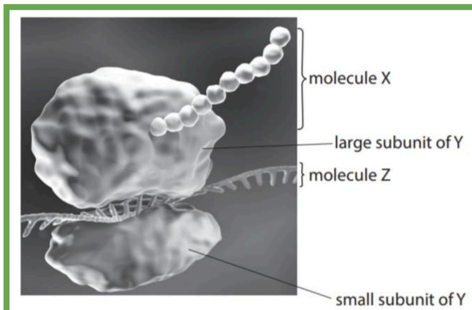
- Describe the role of **one** organelle involved in the production of acrosin (an enzyme).
  - *mitochondria – produce ATP for protein synthesis (of enzyme).*
  - *OR nucleus – site of transcription of enzyme gene / location of gene for enzyme.*
  - *OR ribosomes -site of protein synthesis / where polypeptide is formed.*
  - *OR rER -site of protein synthesis / where polypeptide is formed / formation of 3° structure / packaging protein into (transport) vesicle.*
  - *OR Golgi apparatus – modification of {protein / enzyme} / packaging {protein / enzyme} into (secretory) vesicles.*

➢

Statement	Organelle		
			
absorbs light energy			✓
contains a double membrane		✓	✓
contains ribosomes		✓	✓
forms spindle fibres	✓		

03. Protein Transport:

➢



Molecule X is protein, Molecule Z is RNA, Y is the ribosome.

- (R.Q): Describe the role of the Golgi Apparatus in the formation and transport of extracellular enzymes.
  - *{(rER) vesicles fuse with / protein enters} Golgi where {protein / enzyme} is modified and {protein / enzyme} is packaged into (secretory) vesicles (by Golgi) where the enzyme in these secretory vesicles leave the cell by exocytosis/ vesicles fuse with cell (surface) membrane and release enzyme(s) outside cell.*
- (R.Q): Describe the role of the rough endoplasmic reticulum and the Golgi apparatus in the formation of extracellular enzymes.
- Describe the role of the rough endoplasmic reticulum and the Golgi apparatus in the production of enzymes that prevent polyspermy.
  - *translation occurs at rER ribosomes, where the rER ribosomes join amino acids together forming primary structure/polypeptide chain, and then the {polypeptide (chain) / chain of amino acids} enters rER where the formation of {α-helix / β-pleated sheet / secondary structure / tertiary structure / 3D shape / globular structure} takes place, (causing) formation of {hydrogen bonds / covalent bonds / ionic bonds / disulphide bridges} between the R groups.*
  - *The {polypeptide / protein / enzyme} is packaged into (transport) vesicles by rER to be transported to Golgi apparatus and fuse with it, where the vesicles then become part of it , and the protein enters Golgi apparatus to be modified.*
  - *Modification occurs in Golgi (apparatus) e.g. by adding carbohydrate / phosphate (group) / by glycosylation, and the protein is packaged into vesicles (by Golgi), where it is then pinched off Golgi and transported to cell surface membrane where the vesicles fuse with it, leaving the cell by exocytosis, releasing/secreting the enzyme outside the cell.*

- (c) Cells in the pancreas use amino acids to synthesise polypeptides.
These cells also secrete enzymes and glycoproteins into the small intestine.

  Cells from the pancreas, containing amino acids labelled with a fluorescent marker, were used in an investigation. The fluorescent marker looked green when seen with a special microscope.

  The percentage of green fluorescence inside and outside the cells was measured at the start and after 60 minutes.

  The results of this investigation are shown in the table.

Time / min	Percentage of green fluorescence (%)	
	Inside the cells	Outside the cells
0	100	0
60	38	62

Explain the results of this investigation.
  - *(the percentage of) labelled amino acids {decreases inside the cell / increases outside the cells}, as polypeptide is packaged into vesicles by rER / {(rER) vesicles fuse with / protein enters} Golgi, where {protein is modified / carbohydrate added to protein} (in Golgi)*
  - *{enzyme(s) / (glyco)protein(s)} packaged into (secretory) vesicles (by Golgi) and leave the cell by exocytosis / some (intracellular) {enzyme(s) / (glyco)protein(s)} remain in cells .*
- The sperm are released in bundles with the heads held together by a sticky protein. Suggest how the sticky proteins would have been secreted.

- *packaged into a (secretory) vesicle by the Golgi {apparatus / body} that fuses with cell surface membrane to release them outside of the cell*

04. Observing cells "Microscopy":

Term	Definition
Magnification	*how many times bigger an object appears in an image than in real life / how many times larger the image length is than the actual length*
Resolution	*is a measure of the microscope's ability to distinguish between two points which are close together (on an object)*

- State why a light microscope cannot be used to view these bacteria.
  - *too small (0.5µm) to be viewed (under a light microscope) / magnification (of x14 000-16 000) can't be achieved by a light microscope / resolution would be too low}*
- Explain why the bilayer structure of the cell surface membrane could not be seen using a light microscope.
  - *the two {layers (of the membrane) /membranes} are very close together / there is a very small distance between the {two layers / membranes}, therefore, the resolution of a light microscope is not high enough*
- Explain why an organelle cannot be seen using a light microscope.
  - *the {magnification / resolution} of a light microscope is not high enough (to see this organelle) and higher {magnification / resolution} of electron microscope is needed to see this organelle, as it is very small.*
- Explain why the microscope used for photograph B shows more detail than the microscope used for photograph A.
  - *An electron microscope was used for (photograph) B, so more structures {can be seen / are distinguishable as being separate} due to high resolution.*
- Name the type of microscope used for a photograph of a chloroplast. Give one reason for your answer.
  - *(transmission) electron microscope due to its high resolution / as it allows {the (ultra)structure of the chloroplast to be seen / detailed image}*
- Name the type of microscope used to take this photograph of a pollen grain, and give reasons for your answer.
  - *(scanning) electron microscope due to higher {magnification / level of detail/ resolution} than a light microscope.*

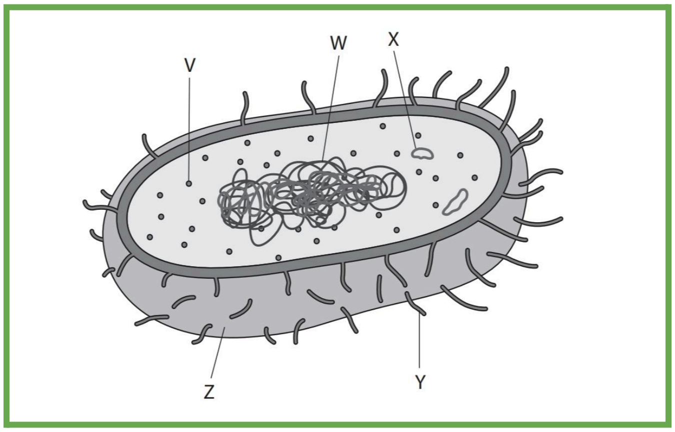
➔ If a photograph using a microscope shows the <u>surface of the structure/the structure from the outside</u> and not its ultrastructure/inner organelles, then a <u>Scanning Electron Microscope (SEM)</u> is used, and ***<u>NOT</u>*** a ~~Transmission Electron Microscope (TEM)~~.

- Give a reason why it is usual to stain specimens in microscopy.
  - *(adding a stain) allows structures to be seen / {differentiate / increase contrast} between structures e.g. chromosomes, DNA, lignin, and organelles and make them visible.*
- State why the cells in this sample were stained before being viewed using a microscope, in order to calculate the mitotic index.
  - *to allow the {chromosomes / chromatids} to be visible / in order to distinguish the phase of mitosis*
- State how a blastocyst cell was treated to show the stage of mitosis.
  - *a {stain / dye} was applied (to the cell) to make the {DNA / chromatin / chromosomes / spindle fibres / histone proteins} visible.*
- Suggest why structures /views of the same organelle can look different in a photograph.
  - *viewed from a different angle / one was transverse and one is longitudinal when the cell was {sliced / cut}.*
  - *different sizes due to different stages of {growth / development}.*

05. Prokaryotic Cells:

Structure	Function
Plasmid	• *contains {genes / DNA / (genetic) information} to make proteins/for antibiotic resistance.* • *enable horizontal transfer of (genetic) information* • *used to exchange (genetic) information with other (prokaryotic) cells*
Circular DNA	• *contains {genes / alleles/ bases/ genetic information} which will be {copied onto mRNA / transcribed / codes for a protein / codes for a sequence of amino acids}*
(Slime) Capsule	• *prevents dehydration of cell* • *has adhesive properties* • *covers bacterial antigens for protection of cell from {white blood cells/ bacteriophages / phagocytes / antibodies / antibiotics/ harsh conditions/ digestion from enzymes / detection by immune system / chemicals}*
Pili	• *allow bacteria to adhere (to surfaces)* • *conjugation / exchange of genetic information / exchange plasmids*
Flagellum	• *Cell motility/ movement (of the cell)*

- Name two structures found in bacterial cells that are involved in the synthesis of enzymes.
  - *(bacterial) chromosome/ nucleoid / (circular) DNA / plasmid / (m)RNA*
  - *(70S) ribosomes.*

➢ 

Y is pilus, Z is Cell wall, V is Ribosome , X is Plasmid, and W is Nucleoid

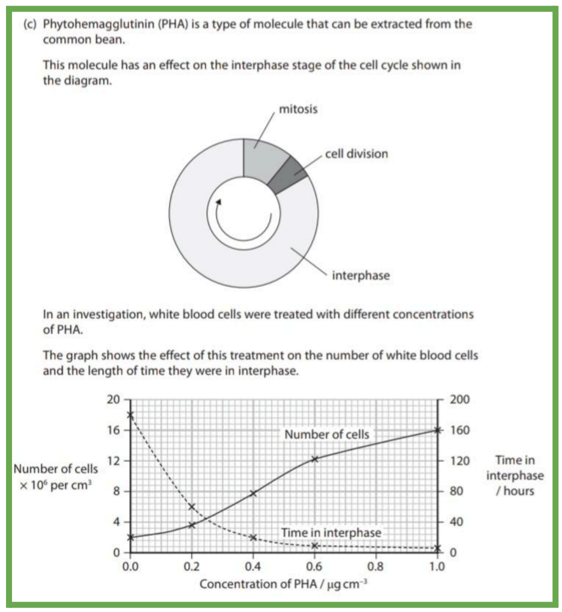
➢

Structure	Animal cell ✓ or X	Plant cell ✓ or X	Prokaryotic cell ✓ or X
amyloplast	X	✓	X
circular DNA	✓	✓	✓
mitochondria	✓	✓	X
nucleolus	✓	✓	X

2. Mitosis, Meiosis, and Reproduction:

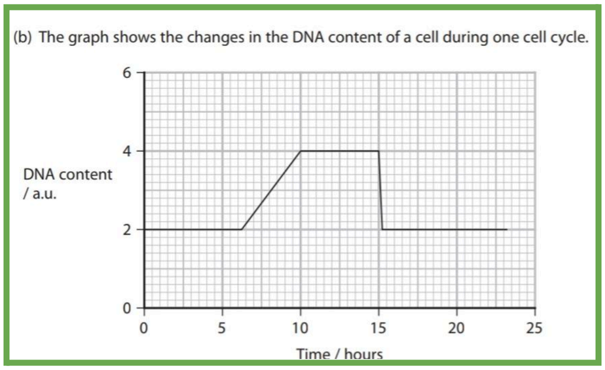
01. The Cell Cycle:

➔ Cell growth, genetic material replication, and organelles synthesis occur during *interphase*.

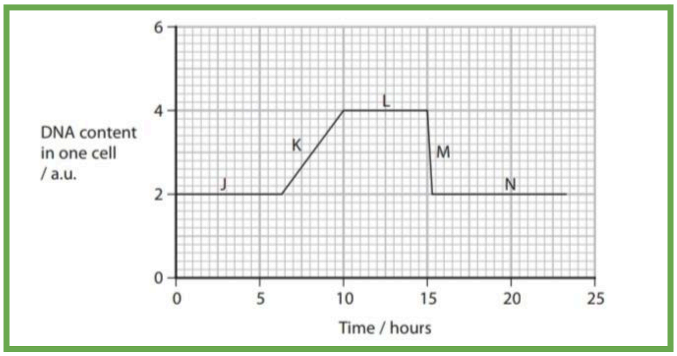
- Describe what happens to DNA during interphase.
    - *(at start of interphase) DNA unwrap from histones / {DNA / chromosomes} {uncoil / uncondensed}*
    - *DNA {is replicated / synthesis / is checked for errors}*
    - *DNA undergoes transcription.*
- Describe what will happen to chromosomes when they enter the interphase stage of the cell cycle.
    - *uncoiling (of chromosome) to form chromatin to be used in protein synthesis / transcription/ DNA replication (in S phase)*
- Methylation of DNA and DNA replication occur in interphase. State one other process that occurs in the nucleus during interphase.
    - *transcription / formation of mRNA/ histone modification / post-transcriptional modification*
- Give two differences in the arrangement of the DNA in a cell at the beginning of interphase and at the end of prophase I.
    - *(at the beginning of) interphase the DNA is {in the nucleus / surrounded by nuclear membrane / surrounded by nuclear envelope} whereas it is not at the end of prophase (I)*
    - *DNA is uncoiled at the beginning of interphase whereas it is condensed at the end of prophase (I)*
    - *DNA has not been replicated at the beginning of interphase whereas it has been replicated at the end of prophase (I)*
- Describe the role of the cell cycle in producing new cells.
    - *synthesis of organelles/ {replication / increasing number} of organelles, synthesis of {proteins / enzymes} required for the next part of the cell cycle and increase in cytoplasm.*
    - *increase in cell size / growth of cell*
    - *{synthesis / replication} of DNA.*
- Suggest why the number and size of the Golgi apparatus change during the cell cycle.
    - *mitosis / two cells formed from one cell, therefore {number / size} of Golgi (in each cell) has to increase (in interphase / G1) in order to provide enough cell {contents / organelles} for two (daughter) cells (after mitosis has occurred), because increased {protein synthesis / protein modification / gene expression} will occur (in interphase) and enzymes/ hormones/ structural protein(s) are required (by the cell).*
- After cell division, the plant cells produced increase in size. Explain how these plant cells increase in size.
    - *because of water uptake / increased {volume of cytoplasm / size of vacuole}*
    - *(increase in size due to) {synthesis/production} of {organelles / (named) proteins / enzymes}, therefore {synthesis/production} of {new/more} cell {membrane / wall}*
- 
(c) Phytohemagglutinin (PHA) is a type of molecule that can be extracted from the common bean.
This molecule has an effect on the interphase stage of the cell cycle shown in the diagram.

In an investigation, white blood cells were treated with different concentrations of PHA.
The graph shows the effect of this treatment on the number of white blood cells and the length of time they were in interphase.

  Explain the results of this investigation.
    - *increase in rate of {mitosis / cell division} / cells proceed to {prophase / mitosis} quicker, (due to) faster {DNA / organelle} replication, (resulting in an) increase in the mitotic index*

- 

  Explain the changes in the DNA content as shown by this graph.
    - *DNA content remains constant during {G1 / G2/ mitosis}*
    - *the DNA content doubles due to {DNA / chromosome} replication / S phase / replication*
    - *the cell divides / cytokinesis (after 15 hours), (therefore) it will produce (two) diploid and genetically identical (daughter) cells.*

➢ 

J and N are G1 phase in Interphase, K is S phase in Interphase, L is G2 in Interphase, and M is Mitosis/Cytokinesis.

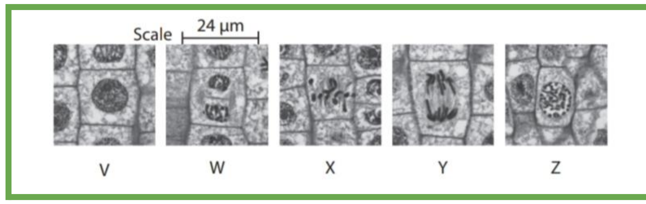
➔ DNA is being replicated at K, and condensation of chromosomes is completed at L (in the graph above)

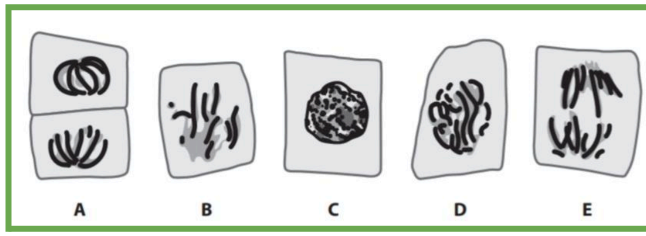
---

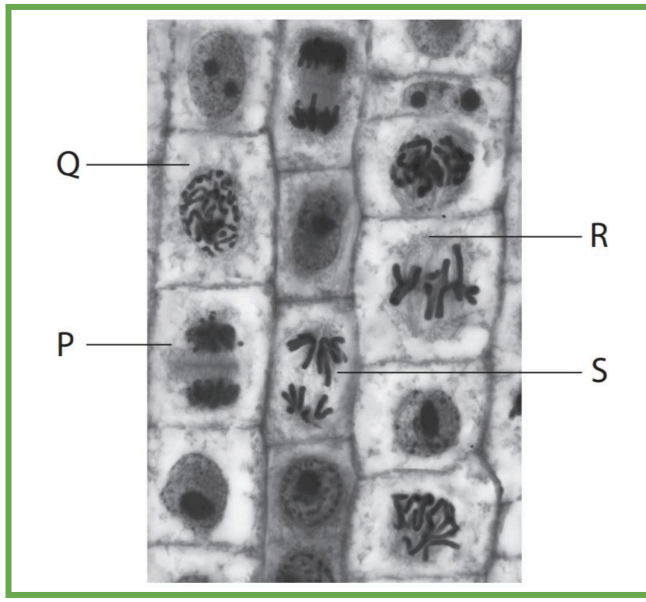
02. Mitosis:

- Give one role, other than for growth or repair, for new cells produced by mitosis.
    - *asexual reproduction*

Stage	Description
Prophase	• *{chromosomes / chromatin / DNA / chromatid} condense* • *nuclear {envelope / membrane} breaks down* • *centrioles {move to poles of the cell / produce spindle fibres/ produce microtubules}*
Metaphase	• *lining up of {chromosomes/sister chromatids} on equator (of cell)* • *centromere bonds to spindle fibres*
Anaphase	• *centromere {splits / divides}* • *{spindle fibres / microtubules} contract* • *one copy of each chromosome pulled to {poles / sides} of cell*
Telophase	• *chromosomes / chromatids / chromatin / DNA} {decondense / uncoil}* • *nuclear {membranes / envelope} reforms around chromosomes* • *spindle fibres {break down / detach from centromere}* • *cell {plate / wall} begins to form* [in plants only]

➢ 

➢ 

➢ 

Stages in order: V and C (Interphase), Z and D and Q (Prophase), X and B and R (Metaphase), Y and E and S (Anaphase), W and A and P (Telophase).

- Compare and contrast the metaphase and the beginning of telophase stages of mitosis.
    - *Similarities:*
        - *both have condensed chromosomes, spindle fibres, and {no nucleus / no nuclear envelope}.*
    - *Differences:*
        - *{spindle fibres / spindle fibres attached to centromere} in metaphase whereas they are beginning to be broken down at beginning of telophase.*
        - *chromosomes are at the {centre / equator} of cell in metaphase whereas they are at poles of cell at beginning of telophase/chromosomes held*

together by centromere in metaphase but not in telophase.
- *condensed chromosomes in metaphase whereas chromosomes beginning to uncondense at beginning of telophase.*
- *no nuclear envelope in metaphase whereas nuclear envelope starting to reform at beginning of telophase.*
- *the cytoplasm starts to divide (cleavage furrow formation) at beginning of telophase whereas it doesn't in metaphase.*

- Explain how preventing the shortening of spindle fibres affects mitosis.
  - *(sister) chromatids cannot be separated / centromere cannot be split, so chromosomes won't {be separated / move to poles of cell / move away from equator} during anaphase, so cells remain in metaphase.*
- Mitotic Index:
  - (R.Q.): State how the mitotic index of a tissue could be calculated.
    - *the number of cells in mitosis divided by the total number of cells (×100)*

$$\text{MITOTIC INDEX} = \frac{\text{NUMBER OF CELLS WITH VISIBLE CHROMOSOMES}}{\text{TOTAL NUMBER OF CELLS}}$$

Formula:

$$\text{Mitotic I} = \frac{(\text{P+M+A+T})}{\text{N}} * 100\%$$

(P+M+A+T) — the sum of all cells in phase as prophase, metaphase, anaphase and telophase, respectively; **N** — total number of cells.

- Explain how preventing the condensation of chromosomes would affect the survival of a patient with cancer.
  - *preventing the condensing of chromosomes would {prevent mitosis occurring / reduce number of cells in mitosis / prevent prophase / prevent metaphase} and (reduced number of cells in mitosis) would {reduce the /lead to a low} mitotic {rate / index}, therefore patients with cancer have a higher percentage survival.*
- Explain why the cells in the morula are genetically identical.
  - *they are all (genetically identical due to being) derived from the **zygote** (cell) , as well as the cells dividing by mitosis which results in genetically identical daughter cells.*
  - *{DNA/ chromosomes} are replicated (in interphase / in semi-conservative replication), therefore, each cell receives one identical copy of each {strand of DNA / chromosome}*

03. Sexual Reproduction and Meiosis:

- Explain how an error in meiosis could result in the production of a zygote with three copies of chromosome 13.
  - *centromere doesn't divide in {anaphase II / meiosis II} , (therefore) both chromosome 13s pulled to same pole of cell / sister chromatids not separated , resulting in {some sperm cells / some nuclei / D} containing two chromosome 13s / {some sperm cells / C} do not have a chromosome 13 , and the {egg cell / female nucleus / female gamete} {contains one chromosome 13 / is haploid}. And, (male and female) nuclei fuse.*
- Suggest a reason why part of chromosome 13 can become attached to a different chromosome when a gamete is formed.
  - *error in crossing over has occurred*
- Suggest how a mutant allele could be produced in prophase I.
  - *(errors in genetic sequence of gene) occur in crossing over, and the chiasmata may be at slightly different points (in the gene)*
- Compare and contrast metaphase in mitosis and meiosis.
  - *Similarities*
    - *lining up of {chromosomes/ chromatids} on equator (of cell)*
    - *centromere bonds to spindle fibres (in metaphase)*
  - *Differences:*
    - *two metaphase stages in meiosis whereas mitosis has one*
    - *{independent/random} assortment occurs (in metaphase I) in meiosis whereas it does not occur in mitosis*
    - *meiosis involves homologous pairs of chromosomes (in metaphase I) and sister chromatids (in metaphase II) whereas mitosis involves sister chromatids/ single chromosomes*
- Explain how meiosis causes genetic variation.
  - *(due to) crossing over, where {sections of chromosomes / sections of chromatids} exchanged at chiasmata between {non-sister / maternal and paternal} chromatids in prophase {I / 1 / one}) results in different allele combinations (in the chromosomes)/results in recombinant chromatids with alleles/genetic information from another (non-sister) chromatid.*
  - *and due to independent assortment, where independent assortment (of chromatids)/(maternal and paternal) chromosomes that occurs (in metaphase I/II) will result in {mixture of (maternal and paternal) chromatids / different combination of alleles on the chromosomes from each parent/ different/new combinations of alleles} in the gametes, in which the genetically different gametes involved (in fertilisation) are random/ in which there is a random {fusion / fertilisation} (of genetically different gametes).*
- Explain how genetically different pollen grains are produced by the same plant.

- *by {meiosis / mutation}, where {crossing over / exchange of alleles} occurs in prophase I, resulting in different combinations of alleles / recombinant {chromosomes / chromatids}/ sister chromatids becoming genetically different from each other, and {independent / random} assortment of maternal and paternal chromosomes occurring in metaphase I.*

- Describe two other processes that can give rise to genetic variation in the offspring of the passion flower, besides mutations and crossing over of non-sister chromatids on homologous chromosomes.
    - *{independent / random} assortment of (maternal and paternal) chromosomes (in metaphase I/II)*
    - *random {fusion / fertilisation} (of genetically different gametes)*
    - *epigenetic modification*

(b) Seed colour in wheat is an example of polygenic inheritance. There are three genes thought to affect seed colour (A, B, C). Each gene has a dominant and recessive allele.

A cross between two heterozygous plants (AaBbCc) was carried out.

The seed colour of the offspring was recorded.

The table below shows the phenotype of the wheat seed colour, the number of recessive alleles and the frequency of seed colour in the offspring.

Colour of wheat seed	Number of recessive alleles	Frequency of seed colour
White	6	1
Very light red	5	6
Light red	4	15
Intermediate red	3	20
Red	2	15
Moderate red	1	6
Dark red	0	1

- Explain this variation in the colour of wheat seeds.
    - *the greater the number of recessive alleles the lighter the colour of the seed / the greater the number of dominant alleles the deeper the red colour.*
    - *random assortment and crossing over results in (gamete) variation, and random fertilisation (of gametes) results in (seed) variation.*
    - *There is a low probability of {inheriting 0/6 recessive alleles / egg cell with no recessive alleles being randomly fertilised by sperm cell with no recessive alleles}*
- Explain how the chromosomes of an egg cell could differ from those in a body cell.
- Explain the differences between the genetic material from an egg cell and a skin cell.
    - *egg cell contains {23 chromosomes/ half the genetic material of the skin cell} / egg cell is haploid (whereas skin/body cell is diploid), (therefore) fertilisation can occur to form a {diploid cell / zygote} and {full set / original number} of chromosomes formed after fertilisation*
    - *the egg cell genetic material/chromosomes is different from the skin/body cell, as they have {an altered base sequence / different alleles} (than body/skin cell chromosomes/genetic material), and that is due to {(random) mutations (during DNA replication), meiosis, crossing over, random/ independent assortment of homologous chromosomes, and/or errors in separation of {chromatids / chromosomes}.*
- Explain how the silver trumpet tree produces seeds that are genetically different from each other.
    - *each male nucleus/gamete and egg cell nucleus/gamete (from silver trumpet trees) is genetically different (from each other) / (each ovule) may have been fertilised by {pollen / gamete} from (many) different trees due to crossing over of {alleles / DNA} (between chromatids) by forming {recombinant chromatids/ different combinations of alleles}/ mutation , and due to {independent / random} assortment (of chromosomes) where both occur in meiosis.*
- Explain why the body cells of the mother and baby have genetic similarities and differences.
    - *(the body cells of) baby will have {half the} chromosomes that are {similar / identical} to those in the mothers (cells, the other {half of} chromosomes in the baby body cells will be (genetically) different as they came from the {sperm cell / father}.*
    - *(there may be genetic differences in the gamete chromosomes inherited from the mother) due to {mutation / crossing over}*

---

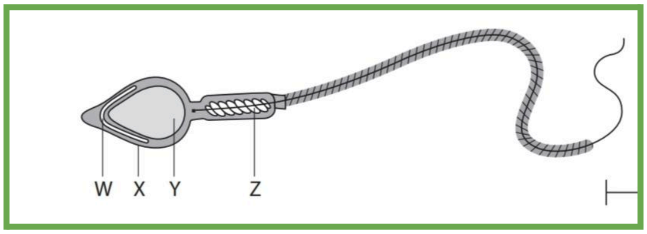
04. Gametes ; Structure and function:

❖ <u>Gametes in Mammals:</u>

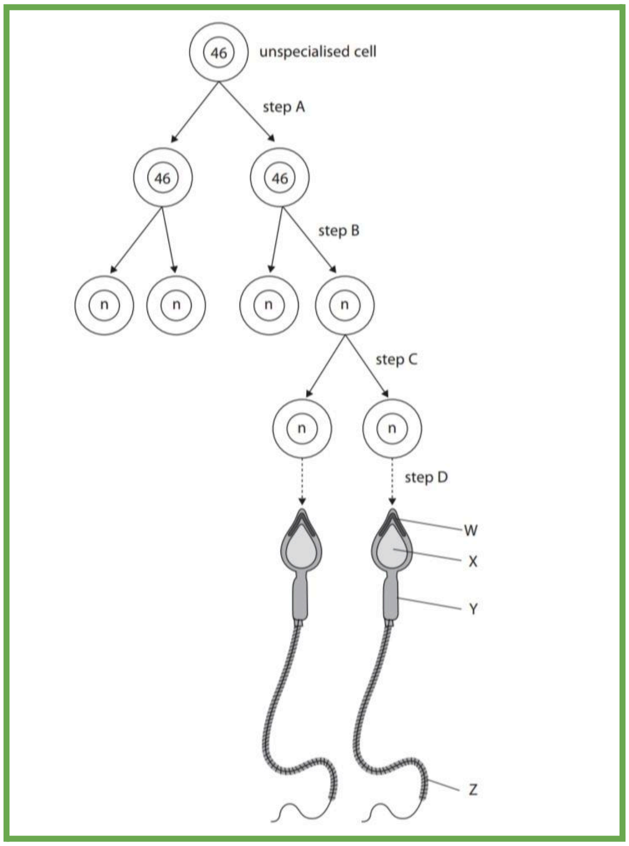
- Sperms:

Adaptations	Function
Head	
Streamlined shape	• *For {efficient movement / speed} and to reduce/decrease {resistance / friction} during movement.*
Haploid nucleus	• *to ensure the zygote {is diploid / has two copies of each chromosome} when an egg cell is fertilized/ to restore diploid number of the zygote.*
Acrosome	• *Fuses with sperm cell membrane, releasing hydrolytic/digestive enzymes which {digest / breakdown} (molecules in) the zona pellucida in order for the sperm cell membrane to fuse with egg cell membrane and thus the nuclei of both gametes fuse in order to form zygote. [Acrosome Reaction]* • *Contains hydrolytic/digestive enzymes (acrosin) which {digest / breakdown} (molecules in) the zona pellucida.*
Antigen/protein	• *on cell surface membrane to bind to receptors on egg cell surface membrane*
Midpiece	

Mitochondria	• *aerobic respiration to {produce / provide} ATP for {movement / motility} of the flagellum/sperm*
Tail	
Flagellum	• *Rotates using energy to allow movement*

➢ 

W is Acrosome, X is Cell membrane, Y is Nucleus, and Z is mitochondria

➢ 

Step A is Mitosis, Steps B and C are Meiosis, Step D is Spermatogenesis.
W is Acrosome, X is Nucleus, Y is Midpiece, and Z is Tail/Flagellum.

- Egg cell (secondary oocyte):

Adaptations	Function
Large cell	• *to store more {lipid/energy}*
Haploid nucleus	• *so that when it is fertilised, ensures {a zygote is formed / it becomes diploid/ zygote has two copies of each chromosome}*
Cortical granules	• *{harden zona pellucida / prevent polyspermy} [cortical reaction]*
Lipid droplets	• Source of energy
Mitochondria	• *{release energy / provide ATP}*
Corona radiata	• *releases chemicals to attract sperm ,*
Zona pellucida	• *Has glycoproteins where sperms bind to*

- Explain why an egg cell has a larger volume than a sperm cell.
    - *contains more cytoplasm (than sperm cell) and endoplasmic reticulum, Golgi apparatus, more mitochondria, ribosomes.*
    - *contains {oil / lipid / triglyceride} (droplets), (which) provides {energy / ATP / monomers} for {mitosis / cell division / development of embryo / growth of embryo}*
    - *Contains {ER / ribosomes / Golgi apparatus} for protein synthesis.*
- Suggest one function of the yolk in the fertilised fish egg cell.
    - *(source of) {fatty acids / glycerol / lipids / amino acids / substrate / glucose / energy / ATP} **for** {respiration / metabolic reactions / protein synthesis / growth of embryo / development of embryo / new cells / mitosis}.*

➢

Structure	Propels male gamete towards female gamete	Modified by the action of cortical granules	Produce ATP by respiration	Contain linear DNA
flagellum	☒ A			
mitochondria			☒ C	
nucleus				☒ D
zona pellucida		☒ B		

❖ Gametes in Plants:

- Pollen grain:

Structure	Function
Tube nucleus (haploid)	• *controls the growth of the pollen tube* • *contains the gene for the production of {digestive / hydrolytic} enzymes (which allows) the (male) {nucleus / nuclei / gamete(s)} to {enter / reach} {ovule / ovary / micropyle / egg cell / fertilise polar nuclei / female gamete}*
Generative nucleus (haploid)	• *(divides to) form two {male / haploid} nuclei* • *one nucleus {fertilizes / fuses with} the egg cell nucleus and one nucleus {fertilizes / fuses with} the polar nuclei*

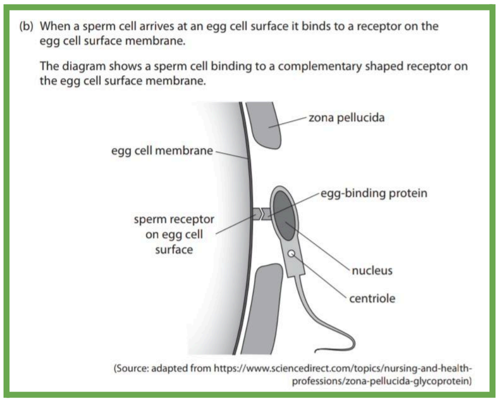
---

05. Fertilisation in mammals and plants:

➔ Which occurs during fertilisation to produce an embryo? *(male nucleus fuses with an egg cell nucleus)*

❖ Fertilisation in Mammals:

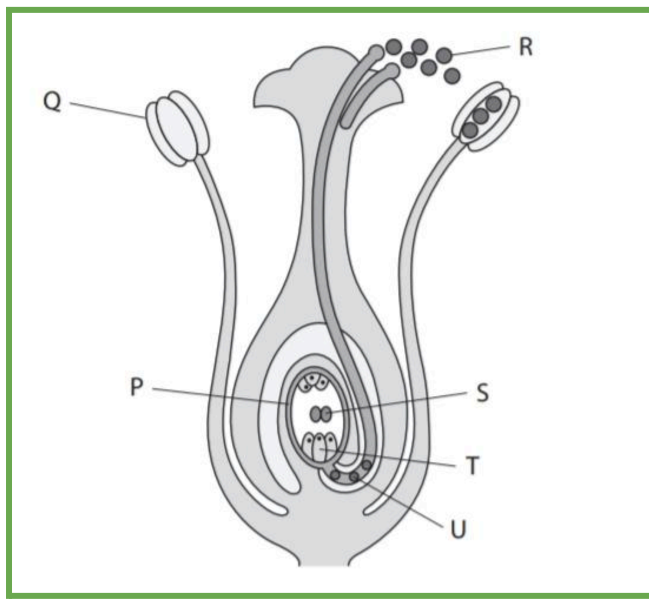
- Describe the acrosome reaction/how the nucleus of a sperm cell enters an egg cell.
    - *acrosome fuses with sperm cell membrane, and {digestive enzymes / hydrolytic enzyme / acrosin} are released , which {digest / breakdown} (molecules in) the zona pellucida , and then sperm cell membrane fuses with {egg cell / oocyte} membrane (allowing the nucleus to enter the egg cell).*



- Suggest why some sperm cells may not be able to bind to the sperm receptor on the egg cell surface membrane.
    - *{ZP3 / egg-binding protein / sperm receptor} may {be missing / not be complementary shape to what it should bind to / changed shape / not be functional}, and the sperm cell {might not have digested through (all) the zona pellucida / may not be able to get through zona pellucida to reach sperm receptor on egg cell membrane}*
    - *Presence of differences in ZP3 / egg-binding protein / receptor due to mutations, epigenetic modification of protein gene.*
- Explain why a higher activity of the digestive enzyme acrosin resulted in a higher percentage of egg cells being fertilised.
    - *increasing the acrosin activity of the sperm cell increases the percentage of egg cells fertilised , (because) higher acrosin activity means the sperm cells can digest through all of the {outer layer / zona pellucida} (of more egg cells), (allowing) sperm (cells) to {bind to (egg cell) membrane / enter egg (cell)}, (so that) sperm nucleus can fuse with egg (cell) nucleus / fusion (of nuclei) can occur.*
    - *Meanwhile, (low acrosin activity) could result in death of sperm cells before fertilisation could occur*
- Explain how cortical granules ensure that the egg cell is diploid after fertilisation.
- Explain the role of the cortical reaction in the process of fertilisation in mammals.
- Describe the events that occur after a sperm cell enters an egg cell, until a zygote is formed.
- Describe what happens when the cortical granules fuse with the cell membrane.
    - *cortical granules are released from the egg cell and fuse with (egg) cell membrane/ zona pellucida, releasing {chemicals / enzymes/ proteases/ proteins} that harden the **zona pellucida** and {removing / blocking / modifying} receptors for sperm , preventing other sperms from binding to the sperm receptors on the zona pellucida. This prevents the entry of more than one sperm cell/polyspermy where {acrosome enzymes / acrosin} can no longer {digest / penetrate} the zona pellucida, ensuring that the nucleus of the zygote that is to be formed will be diploid. Then, the egg cell nuclei and sperm nuclei, which are both haploid, fuse during fertilisation to form a diploid zygote, with a diploid nucleus.*
- Suggest why a fertilised egg cell could have three nuclei.
    - *(when a sperm entered egg cell) cortical granules have fused to cell surface membrane / cortical {reaction/enzymes} resulted in hardening of zona pellucida, and as the zona pellucida (of some egg cells) are damaged there are areas where it is not {present / hardened}, resulting in polyspermy / {an extra / two} sperm have entered (the egg cell).*

➔ A damaged zona pellucida leads to cortical reactions occurring in some areas and not others. The areas that did not have the cortical reaction occurring in them are prone to polyspermy. Also, increasing sperm cell concentration increases the likelihood of polyspermy. [when the Zona Pellucida is damaged]

❖ Fertilisation in Plants:

➢ 

R is Pollen grains, Q is Anther, U is Male (haploid) Nucleus, T is Egg cell, S are Polar Nuclei, and P is Ovule.

- (R.Q): Describe the growth of a pollen tube and the process of fertilisation.
- Describe the events that occur from the time a pollen grain lands on the stigma to the production of a triploid endosperm nucleus and a zygote.
- Describe how the generative nucleus results in the production of an embryo and endosperm tissue in a seed.

- Explain the role of the pollen tube and nuclei in the formation of the endosperm nucleus and fertilisation of flowering plants.
    - *(pollen) tube grows down (style), forming a pathway to reach {ovary / ovule / egg cell / micropyle/ embryo sac} (by the release/secretion of) {digestive / hydrolytic} enzymes used to aid the growth of the tube (to ovule), through the style, and {generative nucleus/male nuclei/ nucleus / nuclei / gamete(s)} are transported to {ovule / ovary / micropyle / egg cell / polar nuclei / female gamete}.*
    - *The generative nucleus {divides / undergoes mitosis} to form two {male / haploid} nuclei, and double fertilisation occurs, where one {male / haploid} nucleus fertilises/fuses with {egg cell / haploid / female}* ***nucleus*** *(to form the diploid zygote/ an embryo), and one {male / haploid} nucleus fertilises/fuses with the two* ***polar {nuclei*** */ bodies} resulting in formation of a {3n/ triploid} endosperm (nucleus).*
- Explain why an embryo cell and an endosperm cell contain different quantities of DNA.
    - *the female gamete(s) and the male nuclei are haploid, and female gamete nucleus was fertilised by {one / a} male {nucleus / gamete} to form the embryo cell, which is {2n / diploid}. Whereas two polar nuclei were fertilised by {one / a} male {nucleus / gamete} to form the endosperm cell, which is {3n / triploid}.*

➔ Pollen grains contain a generative nucleus. The generative nucleus divides to form two male nuclei. *One male nucleus is needed to fertilise an egg cell nucleus to form the embryo.*

---

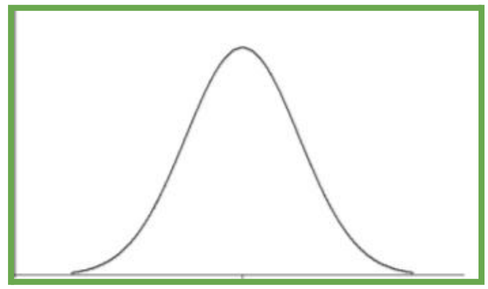
3. Development of Organisms:

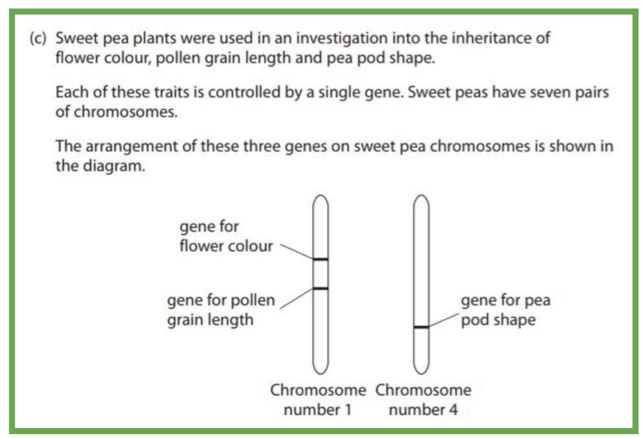
01. Cell differentiation:

Term	Definition
Locus	*{location / position} of a {gene / allele} on a chromosome.*
Allele	*an alternative form / different version of a gene.*
(Genetic) Linkage	*(loci of) {[named]/ specific genes} being on the same {chromosome / chromatid} and so have a higher chance of being inherited together.*
Polygenic inheritance	*(inheritance of) {many/multiple/several/different} genes at different loci that {contribute to / code for} a specific/ the same phenotype/phenotypic {characteristic / trait}*

- Explain how alleles can produce fruit flies with different coloured eyes.
    - *different alleles {have a different DNA base sequence / code for different mRNA}, which may result in a different {polypeptide / protein / pigment} being produced*
    - *The flies may inherit two different alleles / (different allele combinations) may result in {further / intermediate} {phenotypes / eye colours} (in addition to colours in table)*
- Explain how the phenotypes of an individual can show genetic linkage in the inheritance of two genes.
    - *(loci of the genes) are {on same chromosome / close together}, therefore the different versions of the two genes have a higher chance of being {inherited together / not separated by crossing over}*
- Explain why alleles b and e are more likely to be inherited together than alleles A and e.
    - *alleles b and e are closer together , therefore {crossing over / chiasma} is less likely to occur between these alleles.*

➔ The graph that represents polygenic inheritance is a graph of *normal distribution*.





(c) Sweet pea plants were used in an investigation into the inheritance of flower colour, pollen grain length and pea pod shape.

Each of these traits is controlled by a single gene. Sweet peas have seven pairs of chromosomes.

The arrangement of these three genes on sweet pea chromosomes is shown in the diagram.

- Comment on the role of meiosis in the inheritance of these traits.
    - *meiosis increases genetic variation as crossing over and {random/ independent} assortment occur in meiosis.*
    - *The {genes / alleles / loci} for colour and grain length are linked, (therefore alleles for these traits) will be inherited together / (therefore alleles for these traits) unlikely to be separated during crossing over .*
    - *The {genes / alleles / loci} for pea pod wrinkles and {colour/grain length} are not linked , (therefore) the alleles for these traits will be inherited independently due to independent assortment.*

---

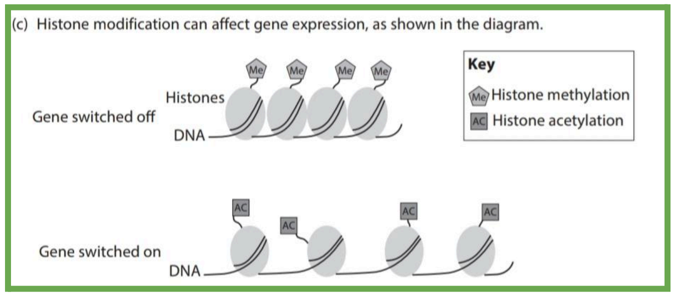
02. Interactions between genes and the environment:

- Suggest two reasons why individuals that have inherited the genotype AaBbCc, may have a darker skin colour than other individuals with the same genotype.
  - *{increased/ more} exposure to {UV/sun} light (would result in a darker skin tone)*
  - *mutation could result in {production of a protein which would result in a darker skin tone / fewer melanocytes}, and epigenetic modification or mutation may have occurred*
  - *limited food source containing materials for pigment synthesis / disease killing pigment cells, such as skin cancer/ disease.*
- A woman with a height of 1.59 m has a daughter. The father is 1.76 m tall. The predicted height for the daughter is 1.61m. Explain why their daughter may not grow to be this height.
  - *{height / phenotype / a characteristic} is affected by the environment, and also depends on the combination of alleles inherited.*
  - *The daughter could be shorter due to {malnutrition / lack of nutrients}, but (could also be taller) due to more nutrients than her parents.*
  - *Diseases, death of child, living in very cold climate, excessive exercise, mutations, and more energy used for heat production all reduce the energy available for growth, leading to a shorter height.*

---

03. Controlling gene expression "epigenetics ":

❖ <u>Gene Expression and Epigenetics:</u>

- Explain why epigenetic changes to genes, such as DNA methylation, are passed on by mitosis.
  - *the DNA methylation is copied when the DNA is replicated / {methylated DNA is / epigenetic changes are} replicated , (therefore) daughter cells will contain the same DNA methylation after cell division / methylated DNA copies are separated in {anaphase / mitosis}*
- Describe how the activity of a gene, such as Rab27a, could be altered.
- When nerve cells are damaged in this starfish, the gene sox2 is activated. Suggest how cell damage could lead to activation of this gene.
  - *stimulation of chemical responses leading to epigenetic modification of {DNA / histones} where where {methyl} group is added to {histone / lysine} or methyl group is {added to} {DNA / cytosine / CpG site} by DNA methyltransferase (DNMT) , therefore DNA is more tightly wrapped around histones/ heterochromatin is formed and genes become {switched off / deactivated / silenced}, thus the {RNA polymerase / transcription factor} cannot bind (to promoter region) OR {acetyl/phosphate} group added to {histone / lysine} or methyl group {removed from} {DNA / cytosine / CpG site}, therefore, DNA is less tightly wrapped around histones / euchromatin is formed and genes become {switched on/ activated/ expressed}, thus the {RNA polymerase / transcription factor} can bind (to promoter region).*
- Explain how DNA methylation could inhibit the production of mRNA.
  - *{methyl / -CH3} groups are added to {cytosine / CpG site} by DNA methyltransferase, so heterochromatin is formed / DNA coiled more tightly around histones, preventing {RNA polymerase / transcription factors} binding to promoter region, therefore preventing transcription.*
- Suggest the role of the enzyme DNA methyltransferase (DNMT) in the process of DNA methylation.
  - *binds {methyl group / CH3 molecule} to DNA/ {cytosine / CpG} (in promoter region)*
- 

  Explain how histone modification can affect gene expression.
  - *methylation is addition of a {methyl / CH3} group to {histone / lysine / arginine} , and histone methylation results in DNA being more tightly wrapped around histones / supercoiling/ {histones/nucleosomes} closer together / formation of heterochromatin}, resulting in the gene being {switched off / silenced / not expressed / not transcribed}, and {RNA polymerase / transcription factors} can't {bind / access} (if gene is switched off).*
  - *acetylation is addition of an {acetyl / COCH3} group to {histone / lysine}, and histone acetylation results in in the {histones /nucleosomes} being further apart / DNA less tightly wrapped around histones / formation of euchromatin, resulting in {RNA polymerase / transcription factors} {binding / accessing} to promoter region (if gene is switched on), and the gene being {switched on / expressed / activated / transcribed}.*
- Histone modification and DNA methylation are processes that occur when totipotent stem cells develop into pluripotent stem cells. Explain how these two processes alter the activation of genes in the cells.
- Describe one form of epigenetic modification that could occur in rabbit cells.
  - *(DNA methylation is the) addition of {methyl / CH3} group(s) to cytosine/CpG site, and(histone modification is the) {methylation} of {histones / lysine}, (resulting in) compacting chromatin / DNA wrapped more tightly around histones/ formation of heterochromatin, and the inhibition of {transcription factor / RNA polymerase} binding to promoter region, (therefore) preventing {transcription / gene expression}.*
- Uncontrolled cell division can result in the formation of a tumour. The RB gene is a tumour suppressor gene. The RB protein produced from this gene inhibits tumour growth. Lung cancer can occur when this

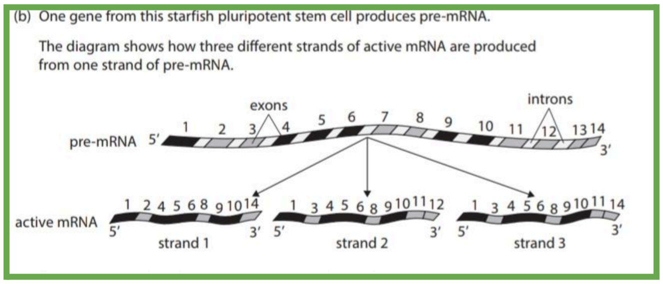
gene has increased methylation compared with the RB gene in normal lung cells. Explain why increased methylation of this gene could cause the growth of a tumour.

- *{methyl groups / CH3} are added to {DNA / the gene/ cytosine} at CpG sites in promoter region, resulting in RB gene being {switched off / silenced}, thus preventing the binding of {RNA polymerase / transcription factors} to promoter region, (therefore) translation (of mRNA) does not occur, and the {RB / tumour suppressor} protein isn't produced, so {the cell cycle / mitosis} continues (forming a tumour) / {reduced / no} inhibition of tumour growth.*

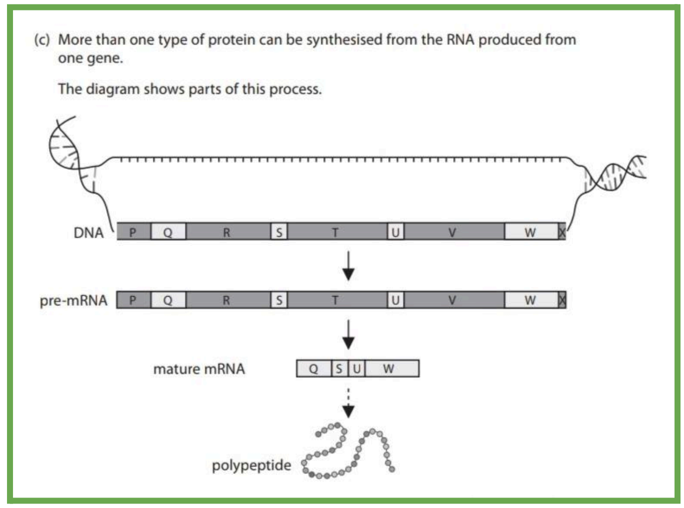
- Explain how tissue samples taken from a parent plant can develop into whole plants that are genetically identical.
  - *tissue sample contains {totipotent / stem / undifferentiated / unspecialised / meristem} cells*
  - *genetically identical because they all contain {the DNA from the parent plant / identical DNA sequence}*
  - *mitosis occurs*
  - *differential gene expression / {epigenetic modification / DNA (de)methylation / histone acetylation / transcription factors} activating genes controlling {root / leaf} cell development , resulting in {differentiation into / growth of} {root / leaf}*
  - *The nutrients in agar: carbohydrates for respiration, nitrates for proteins, calcium ions for calcium pectate, magnesium ions for chlorophyll/ plant hormones.*
- Explain how skeletal muscle cells can contain the same genes as the morula cells but be different in structure and function.
  - *differential gene expression occurred, where only the genes needed for skeletal muscle cell formation are {active / switched on} due to {epigenetic modification}, and only {active / switched on/ skeletal muscle} genes are transcribed (into mRNA). Translation (of the mRNA) occurs (at the ribosome), (resulting in) formation of proteins needed in {skeletal muscle cells / mitochondria / muscle contraction}*
- Explain how a pluripotent stem cell could become a cell that synthesises arginase.
  - *{pluripotent stem cells} can {differentiate / specialise / change into} into {most types of / some types of} cells, through differential gene expression, where the gene for arginase is {active / switched on / not switched off} by {epigenetic modification}, therefore {transcription / translation} occurs (to produce protein)*
- Explain what happens for the cell to become a specialised sperm cell.
  - *differential gene expression occurs, where some genes become {switched on / expressed / switched off}, (due to) epigenetic modification / DNA methylation / histone modification, and so {transcription of / (active) mRNA made from} active genes, therefore translation occurs to form a {polypeptide / protein} which causes {structural/functional} change to cells to change them into (specialised) sperm cell*
- Explain how the cells of the beluga morula change as they develop into the cells of the blastocyst.
  - *morula cells are totipotent whereas blastocyst cells are pluripotent, so the morula cells become pluripotent.*
  - *(specialisation / differentiation and differential gene expression occurs because) gene(s) switched off (in blastocyst cells), due to {epigenetic modification / DNA methylation / histone modification}.*
  - *transcription of (active) genes to produce (active) mRNA and (active) mRNA translated at the ribosomes to produce protein(s) that {determine / change} cell {structure / function}*
- Explain how a mutation could cause the development of teeth in a chicken embryo.
  - *mutation caused differential gene expression that resulted in these genes {becoming switched on / being expressed / not being switched off / remaining switched on}, then {transcription of / (active) mRNA made from} (active tooth production) genes,(therefore) translation (of mRNA) occurs / proteins formed(for teeth development) that cause a structural change to (beak) cells changes them into teeth cells/result in (embryo) cells differentiating into teeth cells.*
- Explain why beta cells can produce insulin but other cells in the pancreas do not.
  - *cell specialisation due to differential gene expression in these cells which causes gene (for insulin production) to be switched off in (other pancreas cells) due to {epigenetic modification / DNA methylation / histone modification}*
  - *Because the gene (for insulin production) {active / switched on / expressed} in these (beta) cells, therefore transcription of (active insulin) gene / (active) mRNA produced only occurs in beta cells, then the translation of (active) mRNA leads to synthesis of insulin protein.*
- Explain how cells in the rabbit embryo become specialised after epigenetic modification.
  - *transcription of active genes / active mRNA produced, therefore the {certain / specific} proteins produced {determine / change} cell {structure / function}*
- Describe how a totipotent stem cell becomes a pluripotent stem cell.
  - ***differential gene expression** where some genes have been (permanently) {inactivated / switched off} by epigenetic modification, and the {proteins / enzymes} {made / synthesised} (from active genes), permanently modify the cell by producing a {structural / functional / metabolic change}*
- Describe how a stem cell in a plant can become a sclerenchyma cell.
  - *differential gene expression, where only some genes are {active / switched on} / some genes are switched off by {epigenetic modification / histone modification / DNA methylation}*
  - *{proteins / enzymes} synthesised (from active genes) which (permanently) modify the cell by producing a {structural / functional / metabolic} change.*

- *modification to become a sclerenchyma cell by synthesis of {cellulose/ microfibrils / lignin} / secondary thickening / lignification of cell walls.*

❖ RNA Splicing:



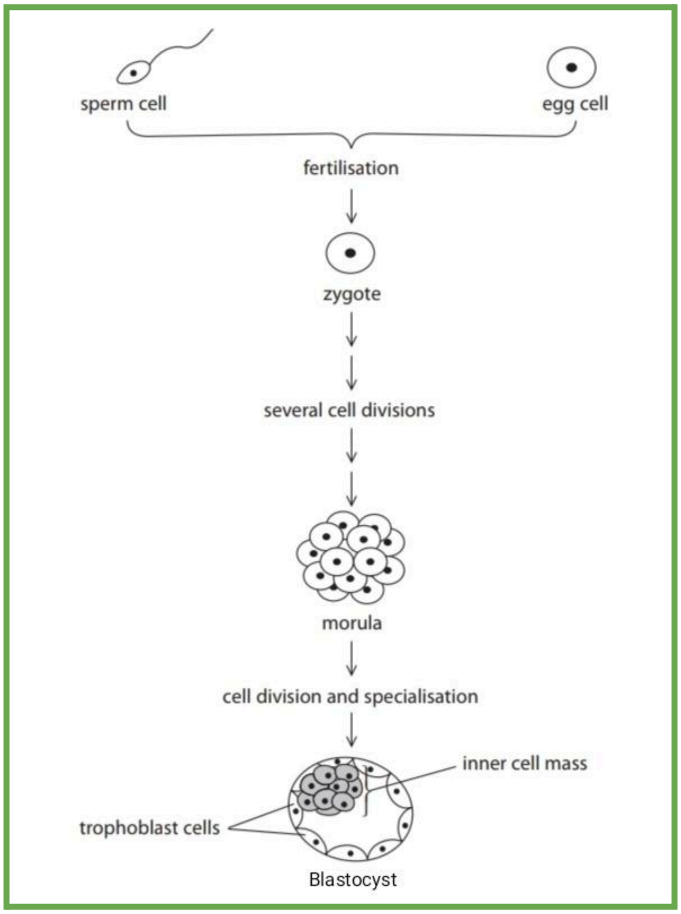
- Explain how these three active mRNA strands are produced.
  - *post-transcriptional {modification/change} (to pre-mRNA), where (all) introns are removed by {enzymes / spliceosomes} and (some) exons removed (to form each strand)*
  - *some strands have had more exons removed / strand 1 contains {fewer / 9} exons / strands 2 and 3 contain {more /10} exons , but all three mRNA strands have {1 / 4 / 5 / 6 / 8 / 9 / 10} (exons)*
- Molecules of pre-mRNA are produced by transcription of active genes. These molecules undergo post-transcriptional changes to form active mRNA. Describe how post-transcriptional changes of pre-mRNA can produce active mRNA.
  - *all introns {cut out / removed} and some exons {cut out / removed} by {spliceosomes / enzymes}, and the exons are joined together to form active mRNA.*
- Explain how the gene responsible for the Dsx-F and Dsx-M proteins can give rise to either a female or a male fly.
  - *post-transcriptional modification (during development of the embryo) / RNA splicing where {introns / non-coding regions} (in Dsx pre-mRNA) removed by {enzymes / spliceosomes} and {rearrangement of / removal of some} {exons / coding regions}, then translation occurs (of active mRNA), (resulting in) a different (Dsx) {primary sequence / sequence of amino acids / polypeptide}.*
  - *The(Dsx-F) proteins result in female specific {cell development / structures / other proteins / cell modification} / converse for males (and Dsx-M)*
- One gene can give rise to more than one type of protein through post-transcriptional changes to the mRNA. One of these changes is the removal of introns. Explain how different proteins can be synthesised from this mRNA that has had introns removed.
  - *some exons can be {cut out / removed} by enzymes/ spliceosomes, and exons can be joined together by enzymes in different orders.*
  - *translation occurs (at ribosome), (therefore) different {amino acid sequences / primary structure / polypeptide chains} are produced.*



- Describe how more than one type of protein can be synthesised from the RNA produced from one gene.
  - *post transcriptional changes / (pre-)RNA splicing, where {introns / P, R, T, V and X} removed by enzymes/ spliceosomes, and the rearrangement of {exons / Q, S, U and W} / removal of some exons*
  - *credit two different permutations of exon order given*
  - *(therefore) a different {primary sequence / sequence of amino acids / polypeptide}*
- Compare and contrast the structures of pre-mRNA and active mRNA.
  - *Similarities*
    - *both are single-stranded and contain the bases A,U,C,G / both contain a base, phosphate group and ribose (sugar) / both contain phosphodiester bonds.*
    - *an exon on pre-mRNA contains same base sequence as the matching exon on active mRNA*
  - *Differences*
    - *pre-mRNA contains both introns and exons whereas active RNA only contains exons/ active mRNA doesn't have introns whereas pre-mRNA does*
    - *pre-mRNA is {longer / has more bases} than active mRNA*
    - *pre-mRNA has recognition site for spliceosome whereas active RNA does not*
    - *active mRNA may have {fewer / different order of} exons (than pre-mRNA)*

---

04. Stem cells:



➢

- Zygote is *totipotent*, but the cells of the inner mass of the blastocyst are *pluripotent*.
- A fertilised egg cell develops into a *morula* and then a *blastocyst*.

- Give one difference between a pluripotent stem cell and a totipotent stem cell.
  - *totipotent (stem cell) can differentiate into {all / more} types of cell including {extra-embryonic / placental} pluripotent cells, whereas pluripotent stem cells can differentiate into {most} types of cells excluding {extra-embryonic / placental} cells.*
- Describe the structure of a blastocyst.
  - *hollow, fluid filled cavity / blastocoel, with an {outer / single} cell layer called trophoblast , and inner cell mass, embryoblast, that contains pluripotent cells (on the inside).*
- Compare and contrast the structure of a morula with the structure of a blastocyst.
  - *Similarities*
    - *both contain diploid cells, dividing by mitosis*
  - *Differences*
    - *morula contains totipotent cells whereas blastocyst contains pluripotent cells*
    - *morula {is a solid ball of / mass of} totipotent cells whereas blastocyst {is a hollow ball of / contains many} pluripotent cells.*
- Describe the role of the morula and the blastocyst in the development of an embryo.
  - *{morula / blastocyst} cells divide by mitosis*
  - *morula cells are totipotent / blastocyst cells are pluripotent*
  - *cells differentiate {from totipotent to pluripotent / into different types of embryonic cells}, where morula cells can differentiate into {placenta / extra embryonic / blastocyst / pluripotent} cells, but blastocyst cells can differentiate into any cells except {placenta / extra embryonic} cells*
  - *inner cell mass becomes the embryo.*

---

05. Using stem cells:

- Suggest why society has approved the use of stem cells, taken from a [named organism]/ [named organism's] embryos.
  - *(stem cells) can {differentiate / specialise} into different {cells / tissues / organs} and so have a great {potential / importance / medical implications} of the research in developing medical therapies, such as {repairing / transplanting} {cells / tissues / organs} in humans, so will save human lives by using them to treat certain diseases like cancer and heart diseases.*
  - *[named organism] does not have a {fully developed nervous system}, and so may not feel pain. Also, [named organism] is not an endangered species, and they are not {harmed / killed} by the process. In addition, they produce a large number of embryos. So, there will be no need to use (human) {embryos /embryonic stem cells}, and so fewer ethical issues.*
- Explain why some people think that research into cancer treatments should not use embryonic stem cells.
  - *(use of embryonic stem cells) means destruction of an embryo (which some people consider to be) an {ethical / moral} issue.*
- Explain why society may support the use of hiPSC stem cells to treat patients with AD.
  - *(hiPSC) are {from skin cells / from (another) adult}, therefore no {destruction / harming} of {embryos / life}*
  - *treatment occurs in {liver/somatic} cells, therefore no changes to {gametes / germ cells}, and treatment {is long lasting / doesn't need to be repeated} .*
- Suggest why injecting pluripotent stem cells may benefit a person who has had a severe heart attack.
  - *pluripotent stem cells have the ability to differentiate into {most cells types / heart (muscle) cells} and so will replace the dead heart (muscle) cells / repair damage caused (to the heart muscle) by the heart attack. so the heart will have improved function.*

---

4. Plant Structure and Function:

01. The Cell wall:

- Cellulose:
  - *contain β-glucose molecules, and 1,4 glycosidic bonds*
  - *forms hydrogen bonds with other cellulose molecules.*

- ◆ *forms microfibrils, and these microfibrils have 1,4 glycosidic , and hydrogen bonds.*

➔ Lignin:
- ◆ *gives waterproofing properties.*
- ◆ *can form spiral shapes.*

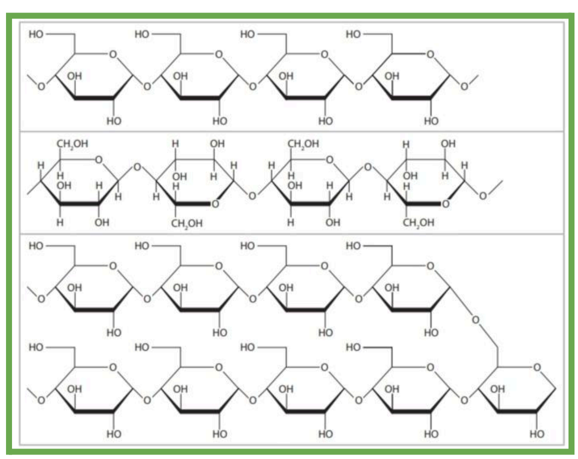
Structure	Function
Thick cell wall	• *made of cellulose fibres support the cell, therefore helps retain rigid structure/ provide support / withstand hydrostatic pressure/ prevent cell lysis.*
Middle lamella	• *joins (adjacent) cell (walls)together, therefore increases {strength /stability} of {plant / cell (wall)}*
Plasmodesta Pits	• *communication between / connects (connected) cells, therefore signalling substances pass through {symplast / cytoplasm} / cytoplasmic streaming.* • *allows transport of minerals, water, glucose, amino acids, proteins, RNA in xylem to adjoining cells/vessels.* • *contribute to lateral movement of water in xylem.*

- Explain how the arrangement of molecules (cellulose and microfibrils) and secondary thickening in the cell walls of xylem vessels and sclerenchyma fibres contributes to their functions (increasing strength of plant cell wall for both tissues, and enabling water transport for xylem)
  - ○ *cellulose molecules can be organised into microfibrils, and many hydrogen bonds form between the (adjacent) {cellulose molecules / microfibrils} that are arranged in {layers / sheets} at different angles (or arranged in a criss-cross pattern/mesh). The microfibril arrangement gives added {tensile strength / rigidity / structural support / stability} (to maintain the structure of the xylem and sclerenchyma) and the presence of (cellulose / microfibrils / hemicellulose / middle lamella/ hydrogen bonds in cellulose molecules/microfibrils / layers (of microfibrils) / lignin / secondary thickening / pectate) for {reduced flexibility* OR increased *strength / support / stability}.*
  - ○ *The {cellulose microfibrils} mesh are held together by/ are embedded in {pectin / hemicellulose}, and presence of lignin in the shape of rings/ spirals/ ladders/ helices in the secondary {cell wall / thickening} (of xylem)* and sclerenchyma *also contribute to the increased strength and stability (for both tissues), lateral movement of water (for xylem), and impermeability of water in the {outside of xylem vessels / cell wall}. The gaps between cellulose microfibrils also allow movement of {water / mineral ions / dissolved solutes} (for xylem)*
- Give three differences between a cell wall with secondary thickening and a cell wall without secondary thickening.
  - ○ *cell wall with secondary thickening {is thicker / has more layers}, as it contains lignin. So, the cell wall with {secondary thickening / lignin} is {stronger / more supported / less flexible / has pits} and is {waterproof / impermeable to water}.*
- Bags made from jute fibres have a high tensile strength. Give one reason why these fibres have a high tensile strength.
  - ○ *due to high {cellulose / lignin} content*
  - ○ *the fibres are {woven into a mesh / twisted with others into a helix} to give a high tensile strength.*
- Suggest two reasons why a climbing plant may have a lower lignin content than a sunflower plant of the same height and diameter.
  - ○ *(it has {support / stability} from {(bamboo) pole / another object} and it (needs to be flexible to) {grow / bend / wraps} around the {(bamboo) pole / another object / supporting object}*

02. Plant organelles:

Structure	Function
Chloroplast	• *site of {photosynthesis / glucose production / light absorption} where it converts light energy to {chemical energy / ATP} to form glucose and oxygen as products of photosynthesis.* • *Site of storage of starch*
Amyloplast	• *Site of storage of starch*
(Sap) Vacuole	• *provides support in which it helps to ensure {turgidity / turgor} / maintains osmotic pressure* • *{stores/dissolves} substances like water, sugars, minerals, proteins, pigments, toxic chemicals, {digestive / hydrolytic} enzymes.*
Tonoplast	• *the membrane of the sap vacuole.*
Starch molecules	• *store/ source of {glucose / energy}*

➔ Amyloplasts, Chloroplasts, Plasmodesma and Golgi Apparatus are all present in plant leaf cells.

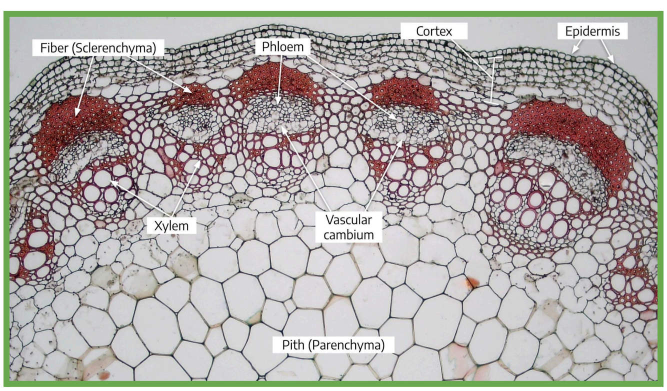
➢ 

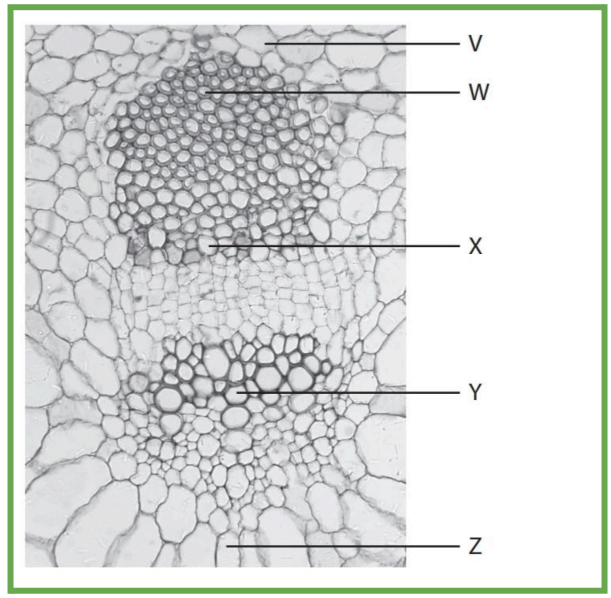
Molecule 1 (at the top) is Amylose, Molecule 2 (in the middle) is Cellulose, and Molecule 3 (at the bottom) is amylopectin

- Describe the structure of starch.
    - *polymer of α-glucose monomers*
    - *composed of amylose and amylopectin*
    - *both 1,4 and 1,6 glycosidic bonds*
    - *{starch / amylopectin} is branched*
- Explain how the structure of starch relates to its function.
    - *{polymer of / contains} glucose because (glucose) is an energy source.*
    - *{branched / 1,6 bonds} therefore can be rapidly hydrolysed (to release glucose/energy).*
    - *compact so more {glucose/energy} can be stored / compact so high energy density}.*
    - *insoluble therefore does not affect osmosis.*
- Compare and contrast the structure of a cellulose molecule and a starch molecule.
    - *Similarity*
        - *both consist of glucose and are {polymers / polysaccharides} containing (1,4) glycosidic bonds*
    - *differences:*
        - *cellulose contains β-glucose (molecules), and 1,4 (glycosidic bonds) and whereas starch contains α-glucose (molecules), and {starch / amylopectin} contains 1,4 and 1,6 (glycosidic bonds).*
        - *cellulose is linear (molecule)/ an unbranched molecule whereas {starch /amylopectin} contains branches.*
- Suggest why the closing of the pores (stoma) reduces water stress in the plant.
    - *to {prevent / reduce} {water loss / transpiration / evaporation} through pore.*

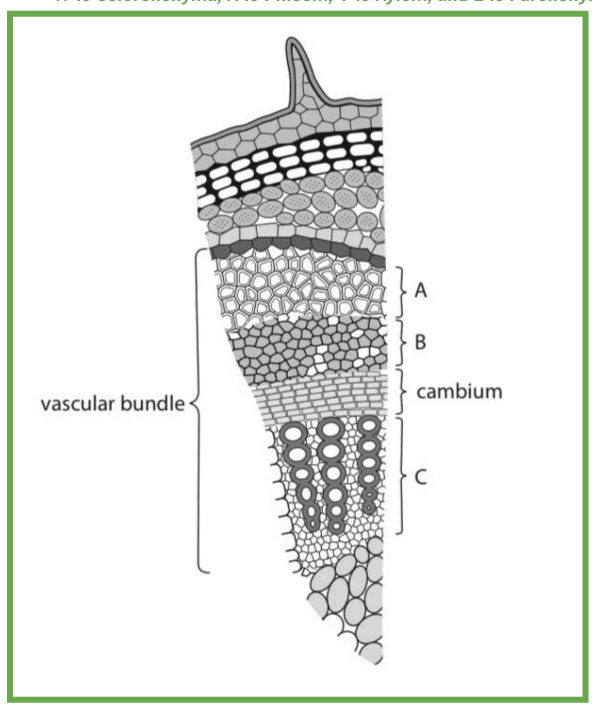
---

03. Plant stems:

➢ 

➢ 

W is Sclerenchyma, X is Phloem, Y is Xylem, and Z is Parenchyma.

➢ 

A is Sclerenchyma, B is Phloem, and C is Xylem.

➔ phloem, sclerenchyma and xylem cell walls *all* contain cellulose.

➔ Fibres from jute plants are used to make bags and string. Give the name of a tissue that would be a source of jute fibres. *(xylem , sclerenchyma , phloem / sieve tube element)*

- Give an example of a tissue that could be produced in a plant.
    - *e.g. xylem, phloem, meristem,*
    - *sclerenchyma, parenchyma, spongy*
    - *mesophyll, palisade mesophyll,*
    - *endosperm.*

Tissue	Function
Sclerenchyma	• *provide support for plant*
Phloem	• *{transport/ translocation} of {sucrose / amino acids/organic solutes} from {source /site of production / leaves} to {sink / site of use / site of storage}.* • *transport of water **down** the stem of plant.*
Xylem	• *transport of water and {mineral ions / inorganic solutes} (to the leaves of the plant)* • *provide support for plant*

- Explain how xylem vessels are adapted for support and transport.
    - *Support*
        - *(cell) walls have secondary thickening / lignin in (cell) wall and (lignin in the cell wall) provides strength and support, and withstanding hydrostatic pressure.*
    - *Transport*
        - *Continuous, hollow tubes with no {end walls / cytoplasm}, to allow transport of {water/mineral ions} with no obstruction to vertical movement / enables formation of continuous water column.*
        - *narrow to aid capillary action.*
        - *contain {pits / non-lignified areas / tracheids} to allow movement of water {in / out} of xylem.*
        - *lignin allows {impermeability to water / waterproofing } to keep water in xylem, and ensures water can only leave through pits.*
- Explain why, after one hour placed in a beaker of stain, the xylem would be a different colour.
    - *because the stain has been {taken up / absorbed / transported} (by xylem / stem), as xylem {transports / absorbs} water (and mineral ions).*

➔ *sclerenchyma* and *xylem* both contain lignin, which gives support to the plant, both have pits, and both have no cytoplasm/are hollow but only the xylem is involved in transport (of water).

- Give one similarity, other than the presence of lignin, and one difference in the structures of sclerenchyma fibres and xylem vessels.
    - *Similarity:*
        - *both contain {cellulose / pits / dead cells / secondary walls}, and {are hollow / do not contain cytoplasm}*
    - *Difference:*
        - *sclerenchyma contain end walls/are {pointed/closed} at each end, whereas xylem do not/are {not / open at each end}.*
        - *xylem contain tracheids whereas sclerenchyma do not.*
- Suggest how the corpse lily plant obtains organic solutes, such as sucrose, from one of the tissues in the vine plant.
    - *from the phloem (of vine plant) / phloem is involved in translocation / phloem is involved in transport of {sucrose / organic solutes}, because part of the corpse lily plant has {grown into the phloem / grown into vascular bundle / made a (physical) connection to the phloem}, and (corpse lily plant absorbs organic solutes / sucrose) by {diffusion / facilitated diffusion / active transport}.*
- Describe how two structures in the phloem enable it to carry out its function.
    - *{(continuous) tube / hollow tube / sieve plate} allows {flow of solution / movement of sucrose and organic solutes / translocation}.*
    - *contain plasmodesmata, so are connected to other cells, allowing {entry /exit} of {sucrose / water / molecules} .*
    - *companion cells to {move sucrose into sieve tube (elements) and provide ATP for active transport}.*
    - *(cellulose) cell wall to withstand pressure.*

➔ Xylem and Phloem:
- ◆ *both xylem vessels and phloem transport dissolved substances*
- ◆ *only phloem transport dissolved substances in both directions "bidirectional"*

- Compare and contrast the structures of phloem sieve tubes and xylem vessels.
    - *Similarities:*
        - *both (fibres) contain cellulose (in the cell wall), have tubular structures, and do not contain a nucleus.*
    - *Differences:*
        - *phloem (sieve tubes) have {sieve plates / (perforated) end walls}, {contain cytoplasm / are not hollow}, contain living cells and no {lignin / secondary thickening}, so are thinner, and have plasmodesmata, whereas xylem (vessels) have no {end walls/ sieve plates}, {do not contain cytoplasm / are hollow} and contain dead cells and {lignin / secondary thickening}, so are thicker, and have pits.*
- Give one difference between the distribution of phloem and xylem in the leaf compared with their distribution in a transverse section of a stem.
    - *xylem and phloem are arranged in a circular pattern in a stem whereas they are arranged in a {horizontal / straight line} in the leaf*
- Give three differences between the distribution of phloem and xylem in the root compared with their distribution in the stem.
    - *the edge of the xylem is closer to the {outside of the structure / epidermis} (than the phloem) in the root / the phloem is closer to the {outside of the structure / epidermis} than the xylem in the stem, and xylem and phloem are contained in (separate vascular) bundles in the stem, separated by cambium.*

- ○ *more equal phloem:xylem in stem (whereas 4:1 in root)*
- ○ *(xylem and phloem / (vascular) bundles) are arranged in {outer part of / a circular pattern in} of stem/ all areas of the stem (in monocots), closer to epidermis.*
- ○ *vascular tissue in one area in the root whereas in several areas in the stem.*

---

04. The importance of water and minerals in plants:

Molecule/Ion	Importance
Water	• *{hydrolysis (reactions) / photosynthesis}.* • *{solvent for / responsible for transport of} inorganic ions / sucrose / assimilates / (in)organic solutes / plant hormones.* • *maintains turgidity of {vacuole / cells}*
Nitrates	• *conversion of glucose* • *formation of {nucleic acid/ amino acids/ nucleotides/ DNA/ RNA} for {(structural) proteins / polypeptides/ enzymes / hormone/ chlorophyll} production, by protein synthesis, which are needed for {growth / repair}.*
Calcium	• *needed to form calcium pectate/pectin, which is needed to {form the middle lamella}, so holds adjacent cells (walls) together and cellulose microfibrils more firmly together (in a matrix), thus increasing strength of the cell wall.*
Magnesium	• *make/form chlorophyll for the production of {glucose / carbohydrate / ATP}/ conversion of absorbed light energy to chemical energy in photosynthesis that occurs.*

- Explain the effect of calcium ion concentration on the firmness of cherry fruits.
  - ○ *increasing the concentration of calcium (ions / chloride) increases (fruit) firmness, (because calcium ions are) needed to make calcium pectate, (forming) middle lamella, and more {calcium pectate / middle lamella} formed would mean the cells were (more) firmly held together/{increases strength of cell wall / holds cells together}*
- Explain the effect EDTA, a chemical that reduces mean mass of calcium ions in the root, would have on new cell walls in the roots of type A maize plants/Explain why Verapamil, a substance that inhibits the uptake of calcium ions by the pollen tube, can reduce the strength of cell walls.
  - ○ *lower concentration of calcium ions (in the cells) means results in reduced {pectin / calcium pectate} molecules formed, therefore increased flexibility of {microfibrils / cellulose} and reduced {strength / stability} of the cell wall (due to fewer {pectin / calcium pectate} molecules).*
- Explain how a lack of magnesium ions could result in yellow leaves and reduced growth.
  - ○ *magnesium (ions) are needed for the formation of {chlorophyll / chloroplasts}, so (yellow leaves are due to) {fewer / no} {chlorophyll (molecules) / chloroplasts / green pigments}, thus reduced growth is due to {reduced / no} {production of glucose / carbohydrate / photosynthesis}.*
- The fungus influenced the expression of certain genes in the plant cells. There was reduced expression of some genes involved in DNA synthesis. There was reduced expression of some genes involved with the synthesis of phospholipids, starch and sucrose. Explain the effect of this fungal infection on the growth of the plant.
  - ○ *reduction in growth due to reduced {cell division / mitosis}/reduced DNA synthesis which results in fewer cells in S phase / not enough DNA produced for mitosis to occur, reduced gene expression which will cause reduction in {enzyme/protein e.g DNA polymerase} production, reduced phospholipids which are needed for (growth of) {cell / organelle} membranes (in daughter cells), and reduced transport of sucrose (around the plant) in the phloem / reduced sucrose available for other plant cells to be used in respiration, so reduced {respiration / ATP production}*

---

05. Using plant starch and fibres:

- Explain what is meant by sustainable.
- Give one reason why plant-based packaging (or any product being plant-based in general) can contribute to sustainability.
  - ○ *The [named] plant is renewable, as it can be regrown, so will be available for future generations and will not run out, thus reduces use of oil-based packaging.*
  - ○ *biodegradable / can be broken down/can decompose by decomposers/microorganisms/bacteria/fungi (more rapidly).*
  - ○ *carbon neutral, so {does not contribute to /reduces} {greenhouse effect / global warming}.*
- Explain why a cup, made from parts of a sugarcane plant that are left over from commercial production of sugar, would be considered a sustainable product.
  - ○ *because {more sugar cane can be grown / there is always more waste from commercial sugar production} , and drinking cups made from plant-based products {can decompose / can be broken down by microorganisms / are biodegradable}*
- Explain what is meant by the term sustainable, with reference to the cutlery produced from the seeds of avocados.
  - ○ *more (cutlery) can (constantly) be made from avocado seeds.*
  - ○ *carbon neutral / lower carbon emissions (than oil-based plastic production) / biodegradable, as avocado seeds are waste products and are {continually being grown / can be grown by future generations}.*
- Explain why a plant-based/starch-based plastic (or any product being plant-based in general) is more sustainable than oil-based plastics/packaging (or any product being oil-based in general).

- *plant-based plastic/packaging is more sustainable (than oil-based plastic), as the plants are renewable and more of them can be regrown, so will be available for future generations and will not run out, thus reducing use of oil-based packaging.*
- *biodegradable / can be broken down/can decompose by decomposers/microorganisms/bacteria/fungi (more rapidly), whereas oil-based plastic/packaging is not.*
- *carbon neutral, so {does not contribute to /reduces} {greenhouse effect / global warming} whereas oil-based plastic/packaging is not.*

- Explain why packaging made from oregano oil, which has antimicrobial properties, may be chosen over traditional oil-based plastic packaging.
  - *plant based packaging is (more) sustainable / biodegradable / carbon neutral (than traditional oil-based packaging), is {renewable / available for future generations}*
  - *(the antimicrobial property of the oregano oil) will {reduce the numbers of (L. monocytogenes) bacteria / prevent bacterial growth (on the food) / prevent microbial growth}, (resulting in) fewer people getting {listeriosis / (microbial) disease} and so food will not go off as quick / extends shelf life.*
- Explain why boards made from oil-based plastics are not considered sustainable.
  - *oil-based plastic is non-renewable, as it is a finite resource / may run out / may not always be available for future generations.*
  - *oil-based plastic {takes a long time to decompose / isn't biodegradable}, so contributes to increased carbon dioxide (in atmosphere) and so {global warming/ greenhouse effect} / not carbon neutral.*
- Suggest one advantage of the vertical system over the horizontal system of growing plants.
  - *{grow/ store} more plants in a given area / less space needed to {grow/ store} the same number of plants, and increases profit as increased yield for lower rent.*

---

06. Plant based medicines:

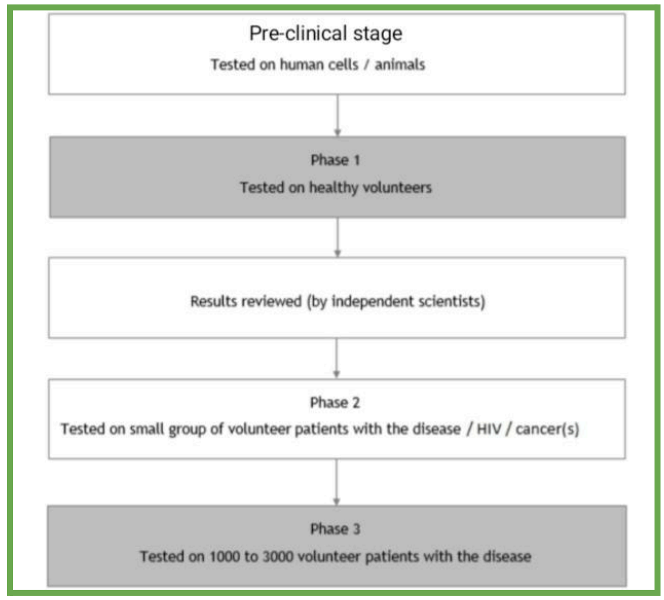
- (R.Q): Explain the conditions that would result in the greatest bacterial growth on the surface of a chopping board.
  - *(some bacteria require) oxygen for (aerobic) respiration, and lack of oxygen for survival of obligate anaerobes.*
  - *glucose for {respiration / ATP production} / amino acids for protein synthesis and growth.*
  - *lipids for synthesis of cell membranes/ to provide fatty acid chains for new cell membranes.*
  - *water for {hydrolysis reactions / solvent / hydration/ forming cytoplasm}.*
  - *optimum temperature and pH for {enzyme-controlled / metabolic} reactions to occur (at suitable or faster rate).*
  - *minerals, such as phosphate, nitrates.*
  - *(suitable) light (intensity) if the bacteria can photosynthesize.*

➔ If a bacteria's growth decreases in the presence of oxygen, then it is an obligate anaerobe / respires anaerobically, and vice versa. The reason why growth decreases is because oxygen would be toxic to it, inhibiting its growth.

- Suggest one reason for the difference in bacterial survival on the pine chopping board and on the plastic chopping board.
  - *pine boards have antimicrobial properties and a pH outside of bacterial optimum pH.*

---

07. Developing new drugs:

➢ 

➔ Plant extracts can also have therapeutic properties. In the 18th century, William Withering tested various extracts made from foxglove plants to treat patients with heart conditions. Name the active ingredient in the extract that William Withering tested. *(digitalis.)*

- Describe how each step in William Withering's method of drug testing compares with contemporary drug testing:
  - Step 1: He guessed that foxglove contained the active ingredient.
    - *modern drugs trials use known active ingredients*
  - Step 2: An extract made from foxgloves was **first** tested on a small number of patients with heart conditions.
    - *extract would first be tested {on a small number of healthy people / in phase I}*
  - Step 3: The dosage and method of extraction were amended after some patients almost died from an overdose.
    - *modern drug trials start with {lowest / smallest} dosage first*

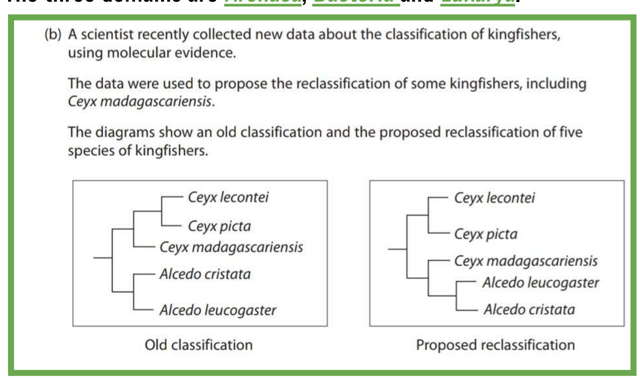
- Step 4: The modified extract was tested on more patients with heart conditions.
  - *modern trials {have three-phased testing / test on larger number of patients / have phase III test}*
- Suggest how a study could be carried out as a double-blind trial.
  - *neither the {patient / group} nor the {doctor / scientist} {would know if they were taking just [named drug 1] or both [named drug 1] and [named drug 2] .*
- Describe a drug trial method that a scientist could use to determine if this extract from L. Candidum is: safe to use in humans and more effective than the current skin burn treatment used in a hospital.
  - *(drug containing L. candidum) extract tested on human(skin) {cells / tissues} with appropriate {dosage / concentrations} (of drug containing L. candidum extract) identified / identification of {side effects / toxicity}.*
  - *(phase 1 / preliminary) tests on (people / volunteers) who {are healthy / do not have skin burns}*
  - *(phase 2) - drug tested on {small / 100 to 500} groups of (patients / people) who have (skin) burns*
  - *(phase 3) - drug tested on {large groups of / 1000 to 3000} (patients / people) who have (skin) burns*
  - *double blind test involves one group receiving extract and the other receiving current (skin burn / hospital) treatment.*
  - *Phase 1 and Phase 3 needed to test for side-effects, and determine {safe / minimum /effective/ suitable/ optimum} {dosage / concentration}/ if it is more effective than {existing treatments / placebo} or adjust the dose.*
- Suggest how a suitable dose for cancer treatment would be determined in human trials.
  - *patients given a range of {doses / concentrations} / doses increase in concentration (throughout the investigation), to (allow selection of) lowest effective concentration / highest dose that can be tolerated without serious side effects/(selection of) dose that is effective with minimal side effects.*
- Describe how a double-blind clinical trial would be performed with this cancer drug.
  - *(selection of) patients with (colon / kidney) cancer, where {half / one group} given {curacin A / (this) cancer drug} and {half / one group} given current (best) drug.*
  - *neither doctor nor patient knows which treatment has been given.*
  - *analysis of data to see if curacin A is more effective than the {current best drug / placebo} and if there is a {significant difference/improvement} between those who took the drug compared to the placebo.*
- Describe how a drug containing flavonoids could be tested in a stage II drug trial.
  - *(flavonoid drug) tested on small numbers of patients {from 100 to 500} with {P.aeruginosa / lung infection} with use of double-blind trial, and placebo / established drug treatment.*
- Describe how a placebo is used in a double-blind trial.
  - *placebo, which has no active ingredient/ is an already known drug / sugar pill / dummy drug, being given to one group (and the other group receiving (anticancer) drug), where neither the patient nor the doctor knows whether the patient is receiving (the drug or) the placebo.*
- The blue-ringed octopus produces a toxin called tetrodoxin. This toxin affects neurones by blocking sodium channels. The pain-relieving properties of this toxin were investigated in a drug trial. Stage II of this drug trial compared the tetrodoxin with the current pain-relieving drug in a group of cancer patients.
  - State the purpose of the current pain-relieving drug in stage II of the drug trial.
    - *Control*
  - Describe the processes that would have occurred before and after stage II in this tetrodoxin drug trial.
    - *(stage I) testing on healthy {people / volunteers} (without cancer)*
    - *(stage III) testing on {1000 to 3000 / large number of} cancer patients*
    - *double blind trial even if in stage II*
    - *analysis of results with (appropriate) statistical test / test for significant difference / review by independent {scientists / medics/FDA} (to see if work can progress to next stage), and identification of appropriate {concentrations / dosage}.*

---

5. Classification:

01. Principles of classification:

- Which domains contain prokaryotic organisms? *Archaea* and *Bacteria*
- If an organism has nucleus, mitochondria, and chloroplasts/ER then it belongs to the *Eukarya* domain.
- The three domains are *Archaea*, *Bacteria* and *Eukarya*.



- Using the proposed reclassification, explain which species of kingfisher are most closely related to Ceyx madagascariensis.
  - *Alcedo leucogaster and Alcedo cristata because they share (the most) recent common ancestor/ share most similarities in their {DNA / named biological molecule}*

Describe the role of the scientific community in evaluating the evidence for this system of classification.
- *peer review*
- *repetition of experiments (by other scientists to see if same data are collected)*
- *{analysis / evaluation / discussion / comparison} of data/ extend the analysis to look for {similarities / differences} in more genes.*

02. What is a species?

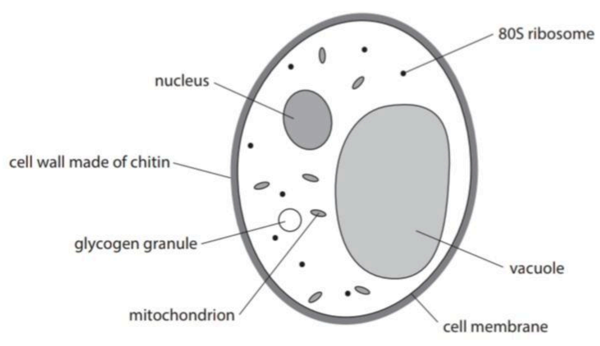
Term	Definition
(R.D): Species	*(a group of) organisms that are capable of {(inter)breeding / mating / reproducing} with others to form fertile offspring.*
Habitat	*place where {organisms / species} live*

- Describe how scientists could determine that this is a new type of bacteria.
  - *analysis of an aspect of phenotype, such as growth characteristics / performing biochemical tests*
  - *analysis of (the sequences in) (circular) DNA, genetic material, proteins, (to see if) there are (many) differences (to existing types of bacteria).*
- Explain how the scientists could confirm that subspecies 1 and subspecies 2 are different species of Archaea.
  - *(analysis of) molecular evidence, for example, DNA, mRNA, proteins, enzymes, to identify similarities and differences in biological molecules / for comparison of biological molecules.*
  - *(analysis of) phenotype, like cell structure / anatomical features, and growth in different {habitats/ conditions} to identify similarities and differences between the two microorganisms (phenotype).*
- Explain how scientists could have determined that subspecies 1 is more closely related to subspecies 2 than to subspecies 3.
  - *analysis of {genotype} by the {comparison / analysis} of the sequences in {DNA/mRNA/amino acid/ gene / allele/ protein} {sequences / structure}, base sequence or intron sequence / proteomics/ molecular phylogeny, and {phenotype} by {comparison/ analysis} of cell structure / anatomical features/ membrane components / cell wall components and growth in different {habitats/ conditions}.*
  - *identification of the number of {similarities / differences} between the two species, and the {more similarities / fewer differences} between molecules means subspecies 1 and subspecies 2 are more closely related/ the subspecies with the {**most** similarities / **fewest** differences} (are the most closely related to subspecies 1.*
- Explain how the zoo determined that its [named] organism was a part of a certain subspecies.
  - *analysis of molecular evidence (of the organism), and then comparison (of molecular evidence) and and {phenotype /anatomy/physiology} to an organism from the subspecies (to see how similar the sequences are)*
  - *Then, breeding the [named] organism in the zoo with an organism from the subspecies to see if **fertile** offspring are produced.*
- Describe the evidence that Carl Woese used to support his system of classification.
  - *(evidence from) molecular phylogeny, and the identification of {similarities / differences} in {DNA / RNA / proteins / enzymes / ribosomes / membrane components / cell wall components}.*
- A scientist suggested that the Akhal-Teke breed is the same as an older breed known as the Turkoman horse. Describe one piece of information that would need to be collected in order to determine if the scientist was correct.
  - *Analysis of {DNA / RNA / gene / allele / amino acid} sequences to see if the sequences from Akhal-Teke and Turkoman breeds are {the same / very similar / few differences}*
- Suggest why the Tristan albatross and wandering albatross were once classified as the same species.
  - *similar phenotype/ they have similar {physical features / anatomy / morphology}, could interbreed to produce fertile offspring.*
- Suggest why two species , who look similar, produce an offspring that would be infertile.
  - *(because the parents are) different species, and {maternal and paternal / 3 and 23}) chromosomes would not pair up, therefore cannot make {haploid / sperm / egg} cells.*
- Suggest why scientists would place organisms into different subspecies of the same species.
  - *same species because they can breed together to form fertile offspring, and different subspecies due to {differences / variation} in {phenotype / characteristics / behaviour / adaptations / molecules / alleles / genome / diet}*
- Give a reason as to why different populations could be described as subspecies and not different species.
  - *individuals (from different sub-species) can (still) breed together to produce fertile offspring.*
- Suggest one way that penguins can be shown to belong to different species.
  - *when penguins from different islands breed together they produce {infertile / sterile} offspring, and presence of differences in {genotype / phenotype}.*

➔ A species can be regrouped from their original taxonomic grouping if there is a presence of {most differences between the DNA sequences / least similarities between the DNA sequences} with other species in the original taxonomic grouping.

03. Domains, Kingdoms or both?

The diagram shows a fungal cell.



Compare and contrast the structure of this fungal cell and a bacterial cell.

- Compare and contrast the structure of this fungal cell and a bacterial cell.
  - *Similarities:*
    - *both contain a cell membrane, cell wall, ribosomes, cytoplasm, glycogen granules, and {DNA / genetic material}.*
  - *Differences:*
    - *differences in cell wall, as cell wall of fungal cell made of chitin while cell wall of bacteria is made of peptidoglycan.*
    - *Presence of linear DNA, nucleus and mitochondria in fungal cell and not in bacterial cell*
    - *Presence of circular DNA, nucleoid, pilus and flagellum in bacterial cell and not fungal cell*
    - *Fungal cell has 80s ribosome while bacterial cell has 70s ribosomes.*

  Compare and contrast the structure of this fungal cell and a plant cell.
  - *similarities:*
    - *both contain membrane bound organelles, such as nucleus, mitochondria, and 80S ribosomes, contain DNA associated with histones, and {cell membrane / cell wall}.*
  - *differences:*
    - *plant cell wall is composed of cellulose and lignin, whereas a fungal cell wall is composed of chitin*
    - *plant cells contain {starch grains / amyloplasts}, {chloroplasts / plasmodesmata / tonoplast}, no lysosomes, and a larger vacuole, whereas fungal cells contain glycogen granules, no {chloroplasts / plasmodesmata / tonoplast}, lysosomes, and a smaller vacuole.*
- A cell is not a prokaryotic cell as it contains chloroplasts. Give one other reason why a cell would not be considered a prokaryotic cell.
  - *the cell has a nucleus, and prokaryotic cells do not contain a nucleus.*
- Give **one** difference between prokaryotic ribosomes and eukaryotic ribosomes/State how ribosomes in prokaryotic cells are different from those in eukaryotic cells.
  - *prokaryotic ribosomes are {70S} and smaller than eukaryotic ribosomes, whereas eukaryotic ribosomes are {80S} and larger than prokaryotic ribosomes.*
- The cells of plants and prokaryotes have cell walls. Name a structural molecule in each type of cell wall.
  - *Plant cell wall: cellulose*
  - *Prokaryotic cell wall: {peptidoglycan / murein}*
- Describe the information the scientists would have used to classify a subspecies into the Archaea domain.
  - *presence of {(peptidoglycan) cell wall / circular chromosome(s) / DNA associated with histones / 70S ribosomes / RNA polymerase}, and ether bond.*
  - *absence of {nucleus / membrane bound organelles}.*

➢

Structure	Archaea only	Bacteria only	Eukarya only	More than one domain
Cell membrane	☒	☒	☒	☒
nucleolus	☒	☒	☒	☒
cell wall	☒	☒	☒	☒
70S (small) ribosome	☒	☒	☒	☒

➢

Feature	Archaea only	Archaea and Bacteria only	Archaea and Eukarya only	Archaea and Bacteria and Eukarya
absence of a nuclear envelope		☒		
presence of circular DNA				☒
presence of a cell membrane				☒
presence of ribosomes				☒

- Some of the bacteria in a river will digest the sewage entering the river. These bacteria will reduce the oxygen content of the river water. Suggest why the oxygen content of the river water is reduced by the bacteria.
  - *bacteria take in oxygen {for aerobic respiration / to produce ATP / to release energy} for asexual reproduction / synthesis of digestive enzymes.*

6. Biodiversity and Conservation:

01. Biodiversity and Endemism:

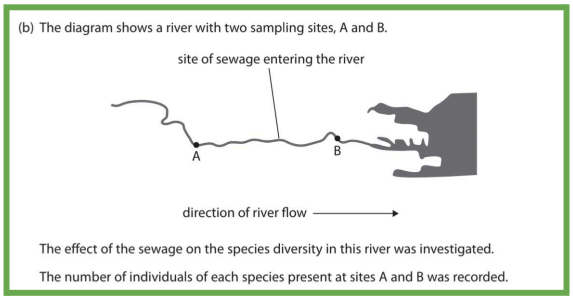
Term	Definition
(R.D): Endemic	*{named plant or species if given/ the species} is only found in {one area / one geographical location / named country if given}*
Species richness	• *number of (different) species in a {defined region / community / specific area / habitat / ecosystem}.* • *{amount / variety} of species in an area.* per unit area

- Suggest what information the scientists would need to collect in order to calculate species richness in a habitat in Yellowstone National Park.

- *(count) number of species*
- *{area / size} of habitat.*

---

02. Measuring Biodiversity:



- Suggest what effect the sewage entering the river will have on the biodiversity at sampling site B. Give a reason for your answer.
  - *decreased {biodiversity / index of diversity} due to loss of species / lower species {richness / abundance} / as more organisms of {one / fewer} species found / lower species evenness*
  - *OR increased {biodiversity / index of diversity}, because new species will be present / higher species richness / presence of species {that are tolerant to low oxygen content / due to more organic material}*
- Explain why the spread of the wild blackberry affects the biodiversity of the island.
  - *reduction in biodiversity, as forest is habitat for many species of plants/ reduction in forest habitat (for many species) / blackberry {outcompetes (the forest) plants / introduces disease}, so populations will decrease (reduction of genetic diversity/gene pool) because of loss of forest that will result in {reduced habitat for animals / reduced food} / increased competition between (animal) species for nesting sites}, and animals not adapted to feed on blackberry / blackberry is poisonous to the animals, so causing reduction in species richness, due to reproduction in {smaller / isolated} populations.*
  - *But there's a possible increase in species that have blackberry as part of food chain.*
- Explain how the reintroduction of beavers resulted in a change in the biodiversity in Scotland.
  - *introduction of new (beaver) species increased the {species richness / biodiversity}, due to {pond being created / new habitat / increased space (due to trees being cut down)} and new food source.*
  - *but caused a reduction in (plant) biodiversity due to trees being cut down / may have caused a reduction in (animal) biodiversity as {habitat / food} reduced (by the trees being cut down)/ {competition with / predation from} beavers*
- Describe how a decrease in biodiversity within an ocean habitat could be determined.
  - *count number of (different) species (in the ocean habitat) / determine species richness, then count number of individuals in each species / determine species {evenness / abundance}*
  - *calculate index of diversity and heterozygosity index / assess genetic diversity*
  - *compare to previous {value / year} to see if there is a decrease*
- The heterozygosity index is a way of measuring the genetic diversity in a population. Write an equation for the heterozygosity index.

$$H = \frac{\text{NUMBER OF HETEROZYGOTES}}{\text{TOTAL POPULATION}}$$

- Describe how the biodiversity of habitats in these areas could be compared.
  - *analyse {the numbers of (different) species (in each habitat)/ species richness} and {the number of individuals in each species / species evenness}*
  - *calculate {index of diversity / diversity index}, then compare {D / index of diversity} values (for habitats in different areas)*
- Write a formula that could be used to determine the biodiversity of a habitat in Yellowstone National Park.

$$D = \frac{N(N-1)}{\Sigma n(n-1)}$$

- The biodiversity of this area of Bhitarkanika National Park could be measured by calculating species richness. Describe one other way in which the biodiversity of this area of Bhitarkanika National Park could be measured.
  - *determining the genetic {diversity / variation} (of a species) by the number of different alleles present and calculating the heterozygosity index*
- Describe how the scientists could calculate the heterozygosity index for a population of Sehuencas water frogs.
  - *{analyse the DNA of (Sehuencas water) frogs / determine the number of heterozygotes} / determine the population size of (Bolivian Sehuencas water) frogs, and then divide the number of heterozygotes by the total number of (Sehuencas water) frogs (in the Bolivia population).*

➔ Weeds on parts of a lake reduces index of diversity, as it blocks light, causing death of plants in the lake, therefore reduction of food/habitats for other organisms.

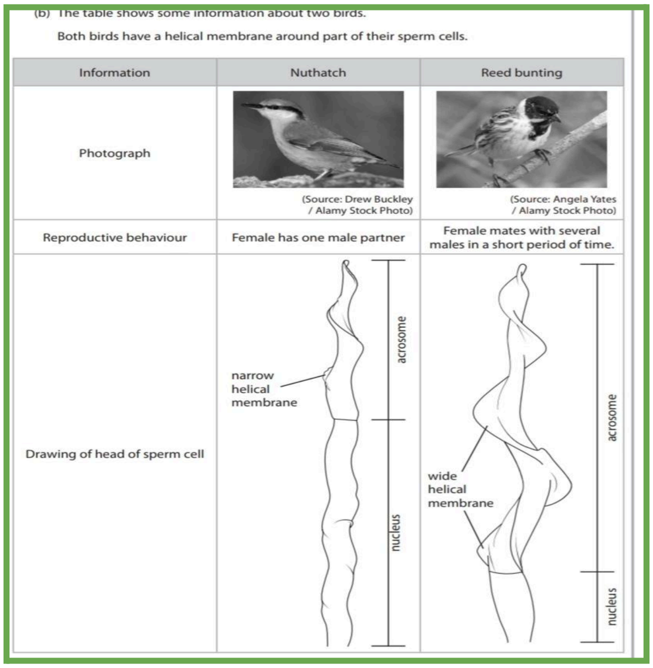
➔ The use of herbicides results in fewer/reduced plant species/diversity, so fewer habitats/shelters for insects, leading to their migration, and reduced food supply for insects, so death of insects due to lack of food.

03. Adaptations to a niche:

Term	Definition
Niche	*role of an organism in its {environment / habitat / ecosystem}*

	Adaptations	Function
BEHAVIOURAL	Making holes in xylem and phloem	• *gains {water for hydrolysis reactions, solvent, and preventing dehydration/ mineral ions} from xylem, and {sucrose for respiration / amino acids for growth} from phloem.*
	Fertilised egg cells kept in mouth until they hatch	• *increases the survival (of fertilised egg cells / offspring) / increase {chance / number} of egg cells hatching by {reducing predation / providing suitable conditions for development}*
	Migration e.g moving higher up the mountains	• *enhances reproductive success, by preventing reproductive isolation, and increases chances of survival.* • *avoiding {predators / hunters / humans / noise pollution}* • *Organism preferring cooler temperatures higher up the mountain* • *{food sources / bamboo / eggs / insects} are found higher up mountain*
ANATOMICAL	Being a hermaphrodite	• *can reproduce with any organism (of the same species) they encounter / in any one encounter both can have fertilised eggs, so an increased number of {offspring produced / population} (in a certain time period).* • *Is a form of (sexual reproduction), so increases genetic {variation / diversity}*
	Hard exoskeleton	• *for {defence / protection} from {predators / damage} / protection of {tissues / organs}*
	Poison (glands)	• *kill {predators / prey} / for {defence / protection / deterrent} from predators/ reduce risk of predation/ reduce risk of being eaten from predators.*
	Streamlined shape / webbed feet / wings	• *enable the organism to swim/ allow it to paddle through water.* • *Webbed feet allows increased air resistance.*
	Long wings	• *to allow it to {fly long distances / glide / use updraft}*
	Long / sharp / big / pointed beak	• *to {allow it to catch/ eat/ feed on its prey / for defence} in the water*
	Small SA: vol	• *to reduce heat loss (in cold water)*
	{thick / lots of} fur	• *to {conserve heat / conserve energy / prevent heat energy loss / act as insulator / protection from cold.*
	Layer of lipid / blubber / feathers	• *for {insulation / thermoregulation}*
	+ large hooves	· not sink in sand

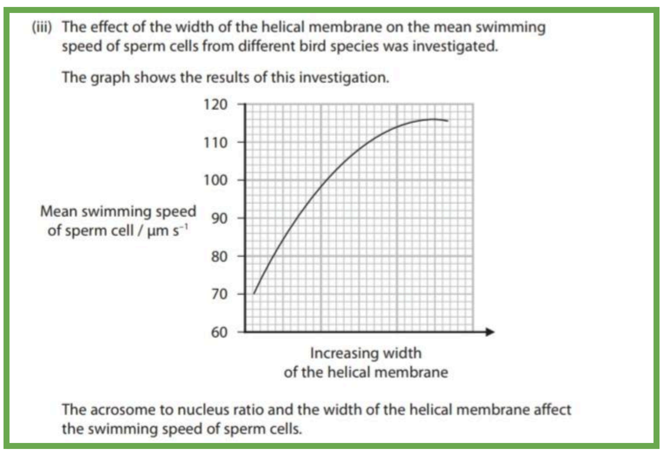
	Adaptations	Function
ANATOMICAL	Colour of feathers / skin related to background / habitat.	• *for camouflage / absorption of heat energy.*
	(coat) {colour / pattern / spots}	• *For camouflage, and deters predators / reduce risk of predation*
	Looking like another organism (mimicry) by having similar shape /size and markings.	• *more likely to catch prey / reduce risk of predation / camouflaged, so less likely to be eaten.* • *predators would avoid venomous prey (IF mimicking a venomous prey)* + deters predators even if organism doesn't produce poison
	Large paws	• *to walk on snow/sand, and increased speed of running on land.* • *For digging*
	Sharp teeth	• *for {biting / tearing / eating / catching} {meat / animals / prey}*
	Sharp / long claws	• *for {added traction / catching prey / climbing (mountains)}* • *For digging.*
	Small ears	• *to reduce heat loss*
	Large tail	• *for {balance / to wrap around to reduce heat energy loss}*
	Short / powerful legs	• *For digging*
	ears face backwards / small eyes	• *So don't get filled with soil when digging*
	Large eyes	• *To see the prey (in the water)*
	Adhesive discs	• *allow the organism to {climb on/ cling to} surfaces (to find food)*
	Enlarged thumbs	• *would help the organism to {climb/grip} trees/branches and {hold / gather / break} {bamboo / fruit / insects / eggs / food / objects}*
PHYSIOLOGICAL	Faster swimming speed of sperms	• *increases chance of fertilising an egg cell (before competitors' sperm do / before death of sperm)*
	Large acrosome of sperms	• *more likely to {have a quicker acrosome reaction / digest the zona pellucida (more) quickly}*
	Long flagellum of sperms	• *faster swimming/ increased force/movement, so increases chance of fertilising an egg cell (before competitors' sperm do / before death of sperm).*
	Tendrils for plants	• *Its coiling {secures / gives support to} the shoot / prevents the shoot from falling over/ allows the plant to climb up other {plants / structures}* • *allows the plant to {grow taller / climb higher}(than other plants)/ divert fewer resources into {strengthening stem / producing supporting tissue (in stem)}* • *So the plant will {outcompete other plants / absorb more light (energy) / carry out more photosynthesis}*

- 

(b) The table shows some information about two birds.

Both birds have a helical membrane around part of their sperm cells.

Information	Nuthatch	Reed bunting
Photograph	(Source: Drew Buckley / Alamy Stock Photo)	(Source: Angela Yates / Alamy Stock Photo)
Reproductive behaviour	Female has one male partner	Female mates with several males in a short period of time.
Drawing of head of sperm cell	narrow helical membrane; acrosome; nucleus	wide helical membrane; acrosome; nucleus



(iii) The effect of the width of the helical membrane on the mean swimming speed of sperm cells from different bird species was investigated.

The graph shows the results of this investigation.

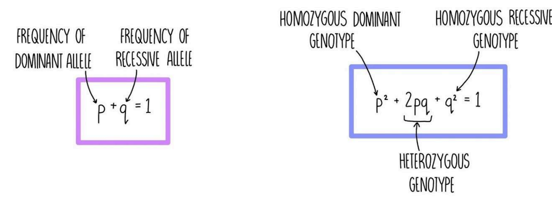
The acrosome to nucleus ratio and the width of the helical membrane affect the swimming speed of sperm cells.

- Explain how these adaptations are related to the reproductive behaviour of these birds.
    - *the (mean) swimming speed (of sperm cells) increases as width of helical membrane increases, as **Reed bunting** males (need sperm cells with faster swimming speeds as they) compete with other males / the female **Reed bunting** mates with more than one male.*
    - *{faster swimming sperm cells / larger helical membranes} are more likely to {fertilise egg cells / form a zygote / have their alleles inherited} (before other males' sperm), and males with larger acrosome (to nucleus ratio are more likely to){have a quicker acrosome reaction / digest the zona pellucida (more) quickly}.*

04. Gene pool and genetic diversity:

- Explain why creating links between subspecies would help maintain genetic diversity.
    - *links would allow individuals to move to other areas / prevent {geographically isolated / reproductively isolated / fragmented} populations, and links between fragmented habitats would enable {outbreeding / individuals from different populations to breed together} , {increasing / allowing} gene flow between populations .*

➔ The Hardy-Weinberg equation can be used to determine changes in allele frequencies over time.



- Explain how scientists could determine if a **change** in the frequency of the recessive allele was occurring over time.
    - *count the number of [named species] and [recessive trait] (of) [named species] in the population/DNA analysis to identify different alleles , and use the Hardy-Weinberg equation to calculate the recessive allele frequency. Then, compare (recessive) allele frequency to recessive allele frequency in previous generation(s) to see if it has changed.*
- Two of the methods used in seed banks are:• store seeds from many different plants of the same species, instead of many seeds from just one plant • regularly germinate samples of the stored seeds, allowing them to grow into adult plants. Explain the advantages of these methods.
    - *(many different plants result in) increased genetic diversity of the seeds stored, because more (seeds from different) {individuals/ plants} increases the probability of having {more alleles / different genotypes / heterozygotes}, and reduces chance of seeds from (mainly) {homozygous recessive / diseased} plants.*
    - *adult plants allow {pollination / sexual reproduction} to occur to produce new seeds, which are genetically different (from parent plants), (for storage), and to replace those seeds stored for a long time/increase the {viability of stored seeds / percentage of seeds that will germinate}.*
- Give two differences between genetic diversity and species richness.
    - *genetic diversity considers one species, one population, and {alleles / genotypes}, whereas species richness considers {different / number of} species, {one habitat / several populations}, {is within a habitat / considers whole organisms / counts number of species (in an area)}.*

05. Reproductive isolation and speciation:

- Explain how a viral disease could result in a change in allele frequencies of these genes in the rabbit population on Lokrum Island.
    - *genetic variation in Lokrum island rabbit population, and the viral disease acts as selection pressure.*
    - *some rabbits had advantageous allele(s) that enabled them to survive (the viral disease) and (these) rabbits*

- *reproduced and passed the advantageous allele(s) on to their offspring, increasing the frequency of the {advantageous / beneficial} allele(s).*
  - *one gene may have greater effect so there will be a greater change in allele frequencies for this gene.*
- Explain how different species of gentoo penguin could have formed on different islands.
  - *the different islands are located a long distance away from each other / the different penguin populations would not meet to breed / no gene flow between populations/ {geographical / allopatric} isolation. Also, there is a presence of {genetic variation / mutations / different alleles} in the penguin population(s).*
  - *Presence of different selection pressure on each island, such as {temperature / climate / predators / prey / environment}.*
  - *{beneficial / advantageous} allele(s) may {give selective advantage in different areas}, meaning those individuals are more likely to survive, and penguins with {beneficial / advantageous} {allele(s) / characteristics} pass these alleles onto their offspring (until the populations become genetically dissimilar to each other).*
  - *(different) populations {have different allele frequencies / different changes in allele frequency}*
  - *the {populations / penguins} on different islands are unable to reproduce to form fertile offspring, as {anatomical / behavioural changes} result in (different species) not being able to mate/ {reproductive / behavioural} isolation.*
- Explain how isolation caused the formation of six fox populations with different characteristics.
  - *each population of island fox cannot mate with individuals on another island, as each island has **different** {(environmental) conditions / habitats / selection pressures}.*
  - *individuals with advantageous **allele** will (survive and) {reproduce/pass onto offspring}, (resulting in) populations being genetically **different** (to each other) / offspring on one island would be genetically different to offspring on a different island / accumulation of different genetic material results in new subspecies.*
- Suggest one selection pressure that results in the development of one of these features: long neck, shell that arches above the neck.
  - *competition for a {mate / territory} and for food.*
- Explain how resistance to this fungus could develop in a population of Sehuencas water frogs.
  - *(genetic) mutation occurred, and (new) allele (coding for protein) involved in fungal resistance.*
  - *frogs resistant to the fungus {survive and breed / have selective advantage}, passing this (resistant) allele onto the offspring, leading to an increase in resistant allele frequency.*
- Suggest how the mice on this island have evolved to become a new species.
  - *(genetic) mutation(s) occurred resulting in (new) allele coding for {larger size / ability to digest meat / eat chicks}, (which then) conferred a selective advantage / (mutated mice) more likely to {survive and reproduce /pass alleles to offspring}, and those without the {mutation /ability to eat chicks / advantageous allele} were less likely to survive and reproduce, therefore increasing allele frequency.*
  - *mutated mice on this island becoming reproductively isolated/ geographically isolated (from non-mutated mice) / allopatric speciation occurred, (resulting in two different species).*
- Suggest how natural selection could have given rise to frogs that can produce poison.
  - *(genetic) mutation(s) occurred, so (new) allele coding for poison production, (which then) conferred a selective advantage / (those with poison) more likely to {survive and reproduce / pass alleles to offspring} and those without the {mutation /poison / advantageous allele} were less likely to survive and reproduce, therefore increasing allele frequency.*
  - *This results in a new species of frog with the ability to produce poison evolved.*

---

06. Conservation; why and how?

❖ <u>Human Activity:</u>

- Suggest why human activity could lead to a species being endangered/extinct.
  - *If it is a tree/plant:*
    - *deforestation / loss of habitat, and increased grazing (of forest)*
    - *reduced number of new trees by reduced {pollination / reproduction / pollinators} / tree doesn't flower often / reduced enzyme activity due to {changing temperatures / drought} / few trees producing seeds / {air pollution / global warming / climate change}.*
    - *disease (that has killed the trees), for example, a fungal disease.*
    - *reduced genetic diversity, by reproducing asexually so no variation.*
  - *If it is an organism:*
    - *habitat loss/ reduction in habitat size/ {isolated pockets of forest / habitat fragmentation/ geographical isolation / loss of food / disruption to food chain / migration / separation of populations} due to {deforestation / urbanisation / water drainage (if the organism lives in water)}*
    - *{pollution / change in salinity} of river making it unsuitable habitat, causing migration or death of the organism. (if the organism lives in water)*
    - *reduction in food due to {habitat loss / competition with other species / pollution/ overfishing (if the organism lives in water)}*
    - *(reduction in population) due to {hunting/poaching by humans / fewer mates /*

disease/ migration due to hunting }, and {gender imbalance / fewer eggs hatching} due to global warming.
        - *introduction of new predator (by humans)*
        - *decreased numbers of prey, so increased competition {for food / territory}.*

❖ Seed Banks:

- (R.Q): Describe and explain how a seed bank would successfully select, prepare and store seeds from trees/plants to aid the conservation of an endangered species.
    - *seeds for this endangered species would be:*
        - *selected from {different plants / different areas / genetically different plants/ varieties} to ensure {different alleles / genetic diversity}.*
        - *{washed / disinfected / sterilised}/treated with an antimicrobial to remove (decomposing) microbes.*
        - *x-rayed to check for {viability / presence of (alive) embryo} so only viable seeds selected.*
        - *prepared by being {dried / dehydrated} to be {viable / stored} for longer/extend storage time of the seeds, prevent/slows down {germination / hydrolysis reactions / metabolic reactions / enzyme reactions / enzyme activation}, and reduce growth of microbes on surface of seed/decomposition.*
        - *stored in suitable conditions, for example , seeds are frozen/seeds placed in temperatures below 0°C/low temperatures/seeds placed in dry conditions to {prevent germination / maintain viability / prevent growth of microbes / reduce enzyme activity / keep them dormant}*
    - *Then, {germination / growth / pollination} of genetically different plants to collect new, replacement seeds from them.*
- Explain why the scientists collected seeds from many individuals of the same plant species.
    - *some seeds may not {be viable / contain embryo / germinate}*
    - *to ensure seed bank contains greater number of different alleles / increase probability of storing all alleles from species / increase probability of not losing alleles from gene pool and to collect representative sample of alleles present in {population / gene pool}*
    - *to ensure alleles that may confer an advantageous trait are stored.*
- Explain why some seed banks would choose to use cryopreservation.
    - *as this may not destroy the embryo / {prevent extinction of / conserve} endangered species that cannot survive seed drying*
    - *prevents {germination when stored / cell division / hydrolysis reactions / metabolic reactions / enzyme activation}*
    - *to increase the {number of different species / genetic diversity / number of alleles} stored in the seed bank*
    - *storing embryos takes up less space than storing seeds*
- Suggest two reasons why seed banks store seeds instead of growing whole plants
    - *can store more seeds as they {take up less space / are smaller}, as growing plants takes up more space.*
    - *more seeds stored aids maintenance of genetic diversity*
    - *seeds need less {maintenance / cost}, and can {survive longer than plants / be frozen }/ can be stored for longer.*

❖ Conservation Programmes and Education:

- Suggest why the population of the St Helena plover decreased after the introduction of cats and rats to the island.
    - *{cats / rats} {ate / destroyed / preyed on} the {eggs / chicks / plovers}*
    - *few(er) chicks to {reproduce / survive to breeding age} / fewer birds to reproduce (next year)*
    - *increased competition for food / {rats / cats} carried diseases which killed the birds*
- Explain how a captive breeding programme/ conservation could maintain the genetic diversity of a species.
    - *For conservation:*
        - *establish protected areas / increase food source.*
        - *reduce the population of {(named) predators}.*
        - *(reduce deaths from disease by) {treating diseased foxes (with medication) / vaccination}.*
    - *For captive breeding programmes:*
        - *use of captive breeding programme (to increase population size) / reintroduce (captive bred) foxes.*
        - *analyse {individual's alleles / gene pool/ genotype of the species}/ identify individuals with different alleles, and use a stud book to ensure genetic diversity of the species' population is maintained.*
        - *breed {individuals} {with different alleles / from different populations}, therefore, preventing loss of alleles (from gene pool) .*
- Describe how St Helena breeding programme could have been carried out.
    - *consideration of analysis of {alleles / gene pool}, then taking pollen from one gumwood tree and using it to pollinate another gumwood tree.*
    - *planting of {seeds / young saplings / shoots} {where they won't be grazed by rabbits / in a safe area / protected area / area with no rabbits}.*
- Explain how scientists could breed new high-yielding varieties of eggplants which are resistant to R. solanacearum bacteria.
    - *set up a {breeding programme / stud book} and {cross / breed} variety A with variety D, because D has {lowest wilt percentage / lowest disease index / best resistance} and variety A has the highest yield / A and D will produce offspring with resistance and high yield.*
    - *check to see if offspring are resistant by infecting grown offspring with the bacterium.*

- *repeat using offspring with required characteristics (to get a new high-yielding and resistant variety)*

- Explain why scientists are breeding the Cavendish banana with different varieties of banana.
    - *(all Cavendish plants) are susceptible to the fungus because they are {genetically identical / clones}, whereas different varieties may have resistance to the fungus / contain an allele for resistance (to fungus), therefore breeding could increase genetic {diversity / variation}, resulting in (new banana) plants with resistance to {fungus / Panama disease}*
- Explain one way that young trees could be protected from herbivorous organisms.
    - *build {fences / barriers} around young trees / spray trees with chemical to deter the organisms/ to prevent the organisms from being able to eat {trees / shoots}*
    - *Or {kill the herbivorous organism / reduce its population / introduce predators } to reduce risk of young trees being eaten / fewer {trees / shoots} are eaten .*
- Describe how education could help conserve an endangered subspecies.
    - *raise awareness of:*
        - *importance of {tigers / biodiversity} / raise awareness {of risk of extinction / that tigers are endangered}*
        - *{protection / conservation / expansion / preventing destruction} of current {range / habitat}*
        - *{develop / increase} breeding programmes / reintroduction of captive bred tigers*
        - *need to decrease poaching / make {hunting / selling tiger parts / buying tiger parts} illegal*
    - *encourage donations for conservation projects.*

I. Statistical Analysis:

- Give the name of a statistical test that could be used to assess the strength of the relationship between length of midpiece and swimming speed of sperm cells.
    - *correlation coefficient / Spearmans rank / Pearsons.*
- *Chi-Squared* test can be used to analyse observed phenotype frequencies and expected phenotype frequencies for the foxglove plants.

Your Own Extra Notes

★ How to Calculate actual size?
-image size divided by magnification